

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

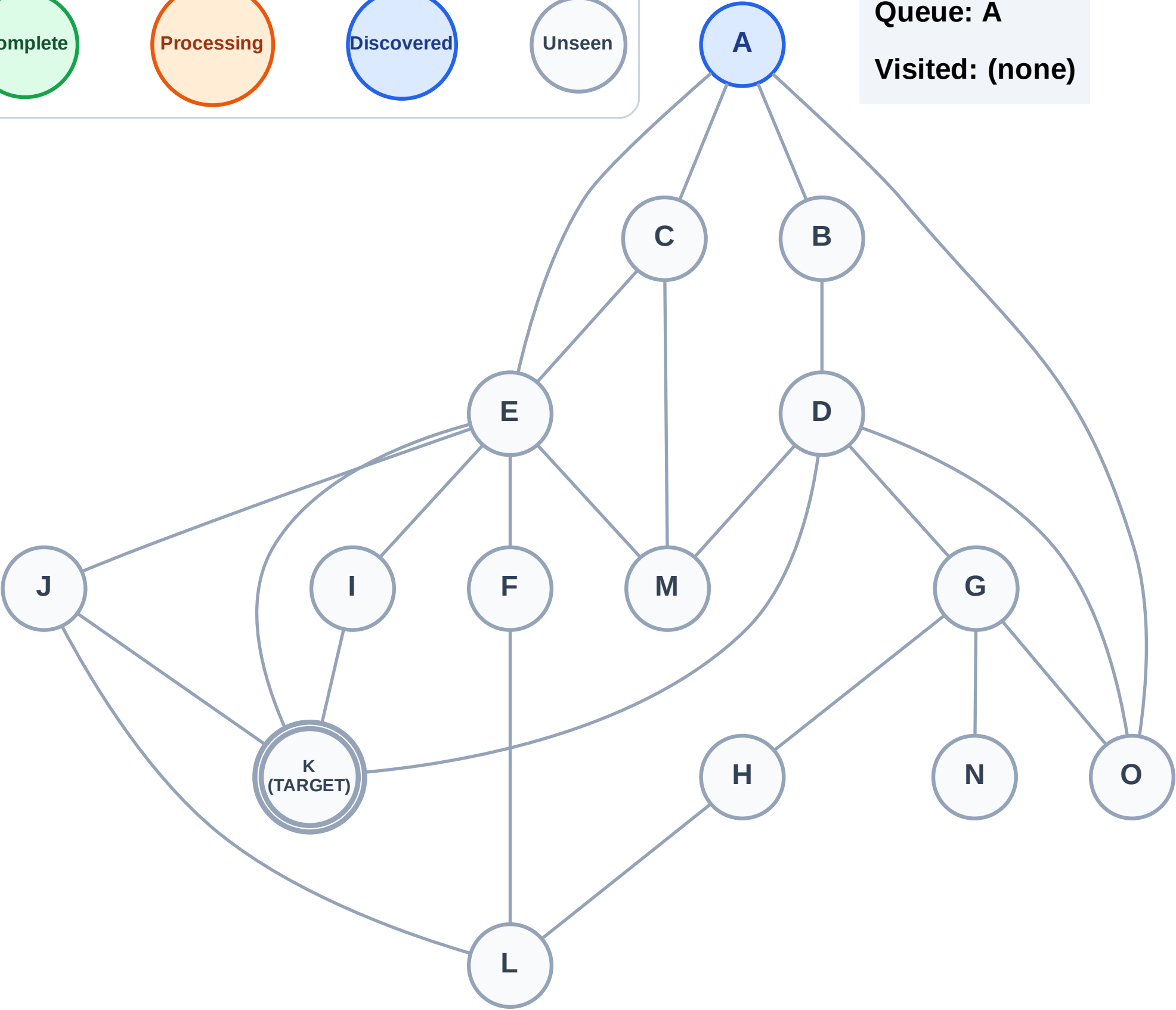
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

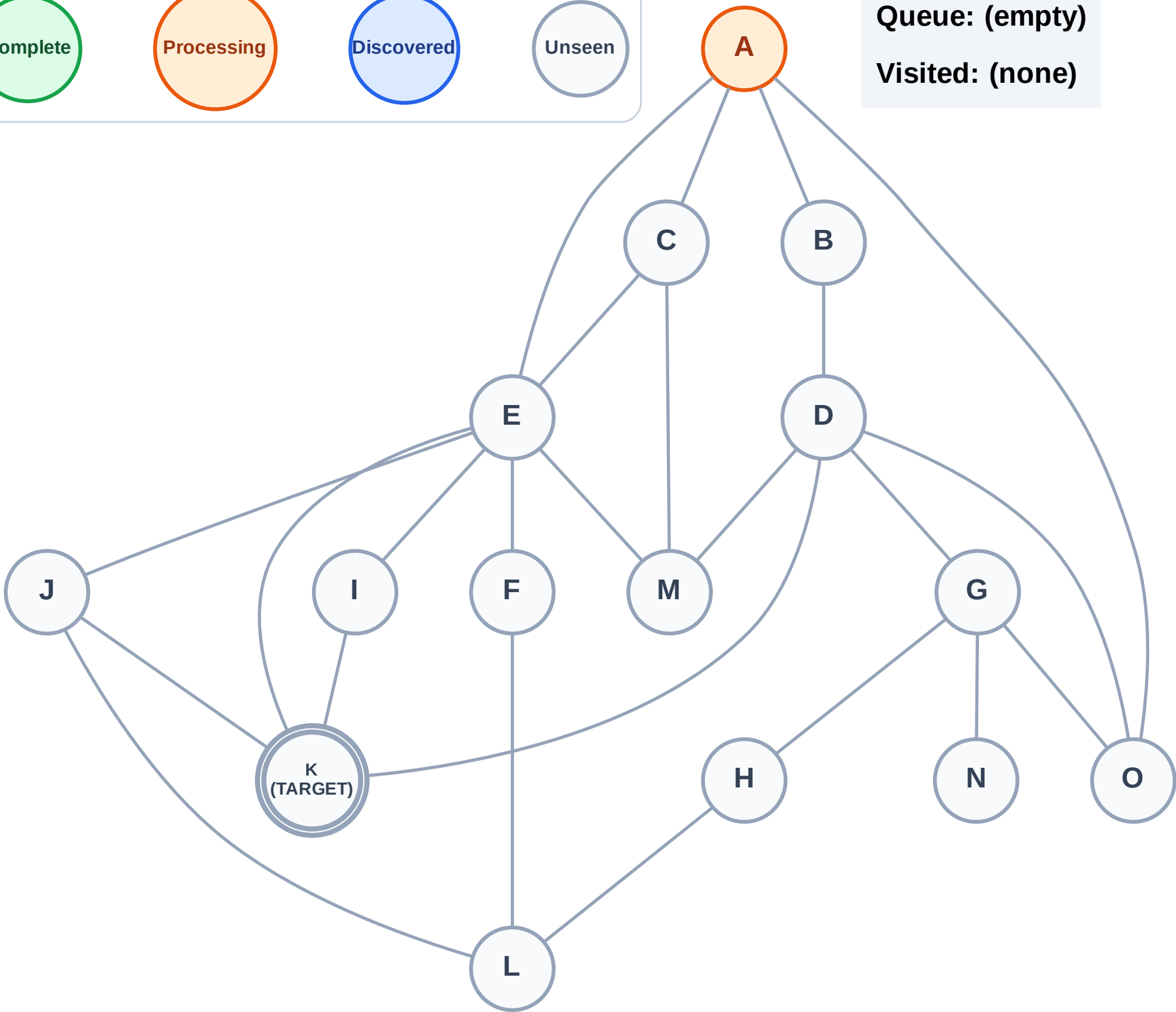
Step 2: Process node A

Legend



Queue: (empty)

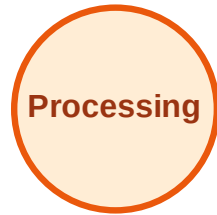
Visited: (none)



BFS Traversal

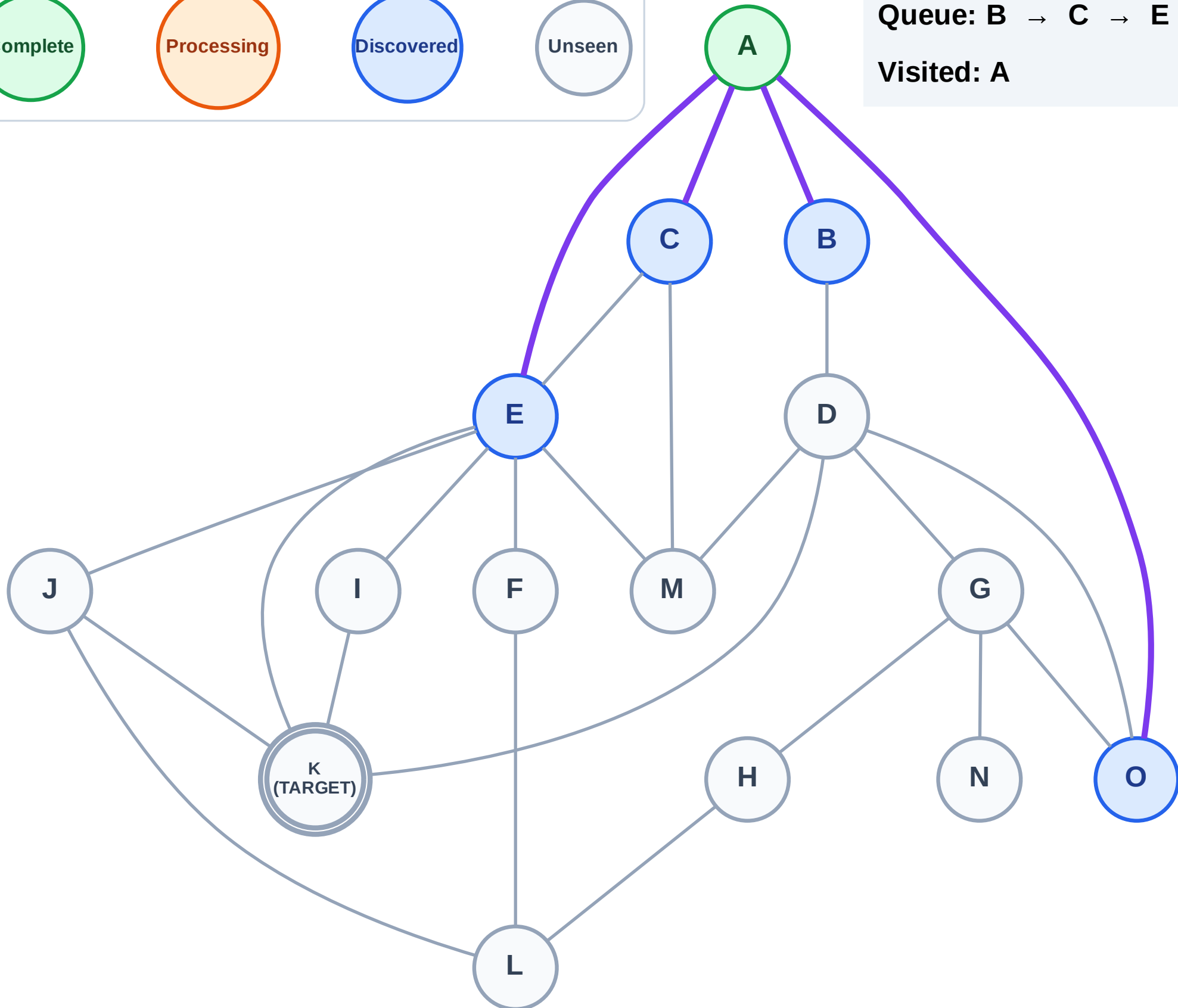
Step 3: Discover B, C, E, O from A

Legend



Queue: B → C → E → O

Visited: A



BFS Traversal

Step 4: Process node B

Queue: C → E → O
Visited: A

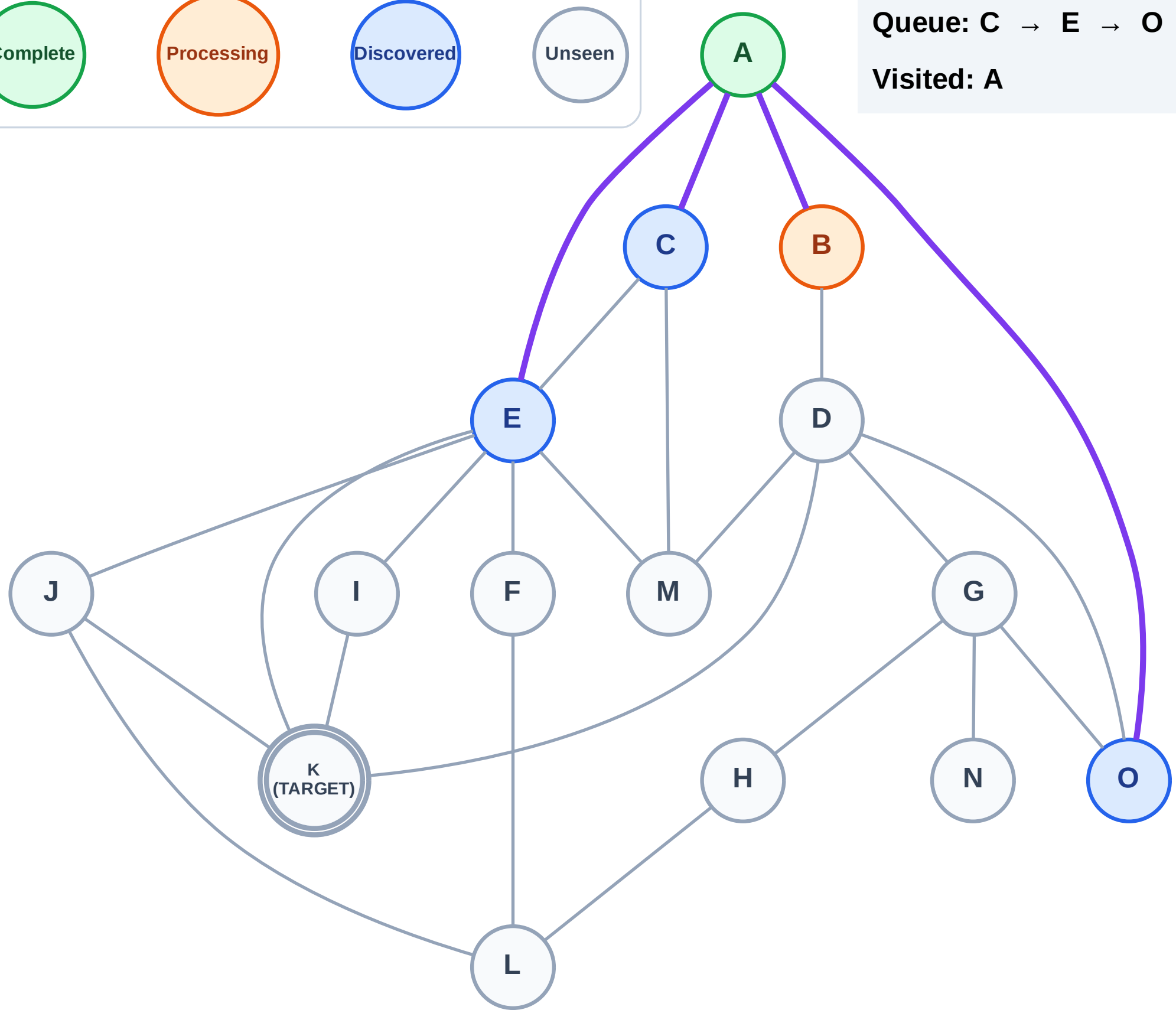
Legend

Complete

Processing

Discovered

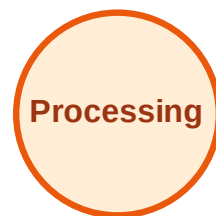
Unseen



BFS Traversal

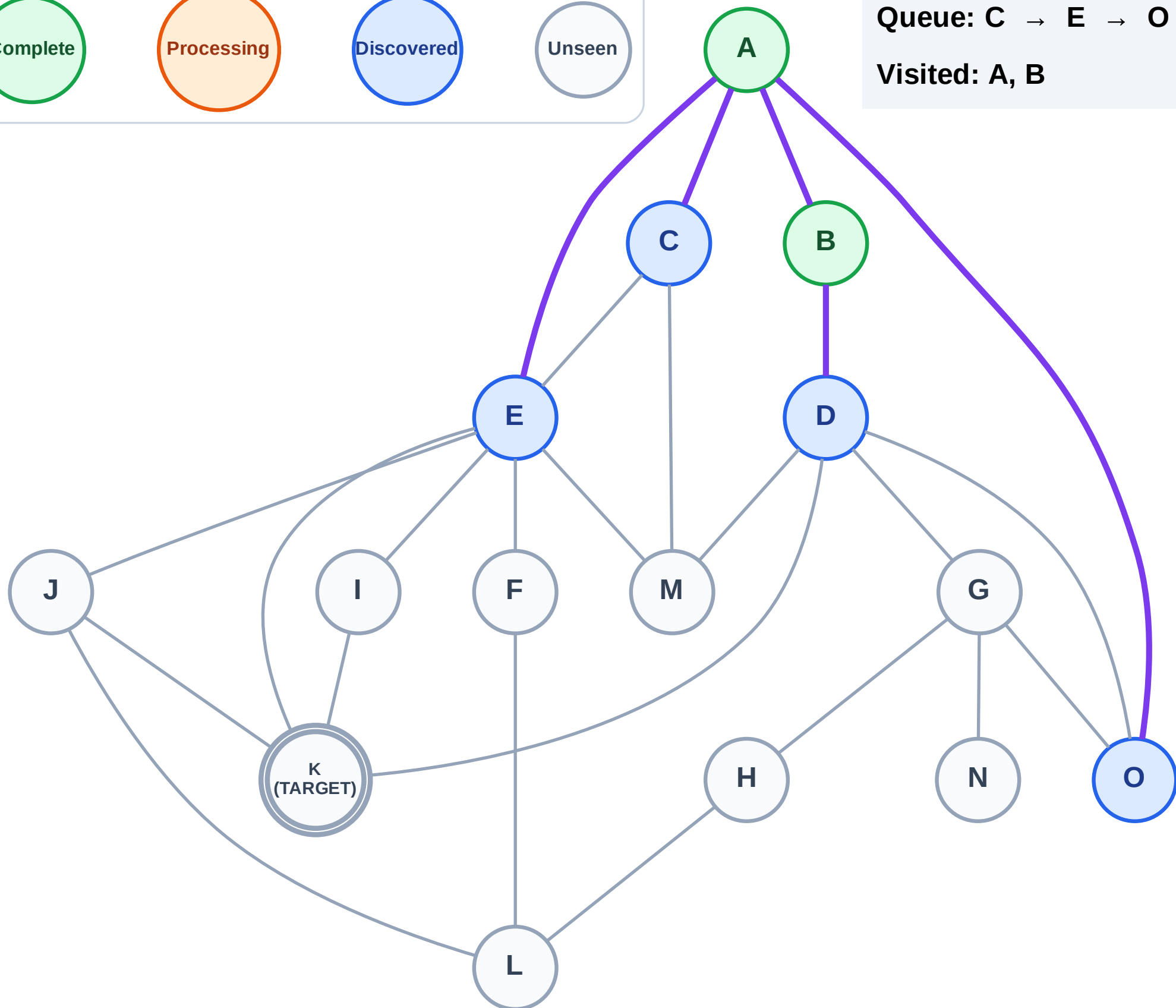
Step 5: Discover D from B

Legend



Queue: C → E → O → D

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

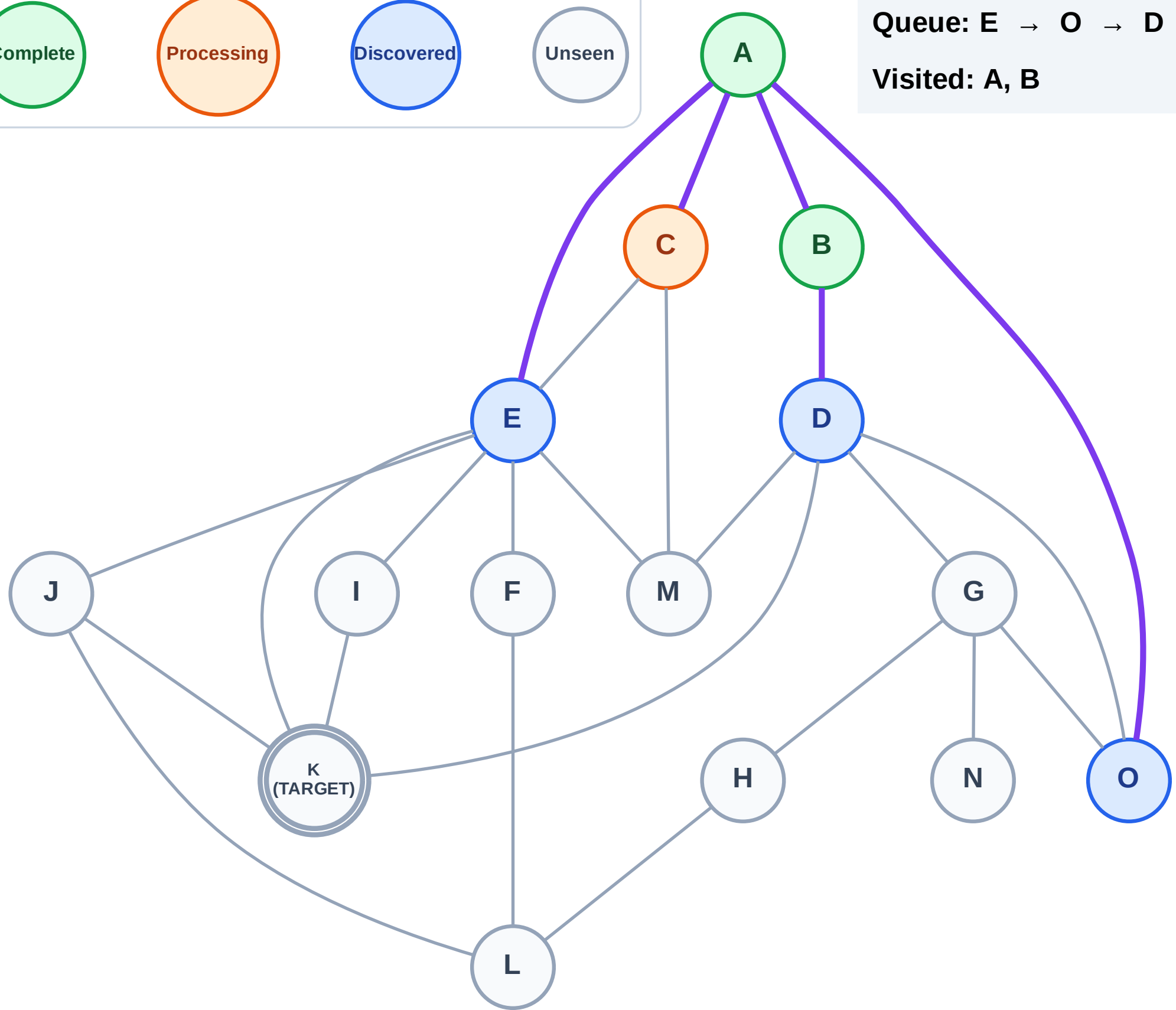
Complete

Processing

Discovered

Unseen

Queue: E → O → D
Visited: A, B

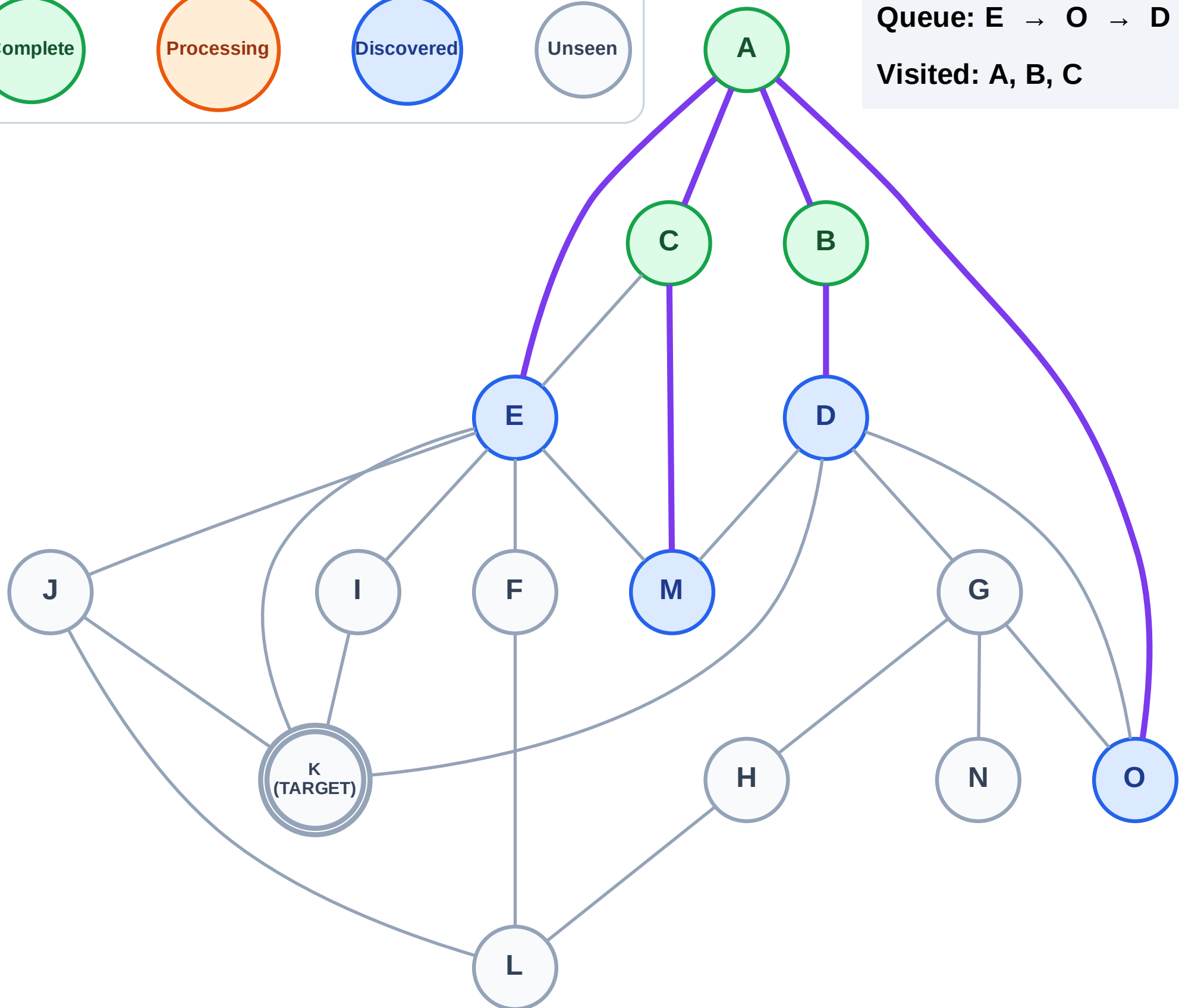


Step 7: Discover M from C

Legend

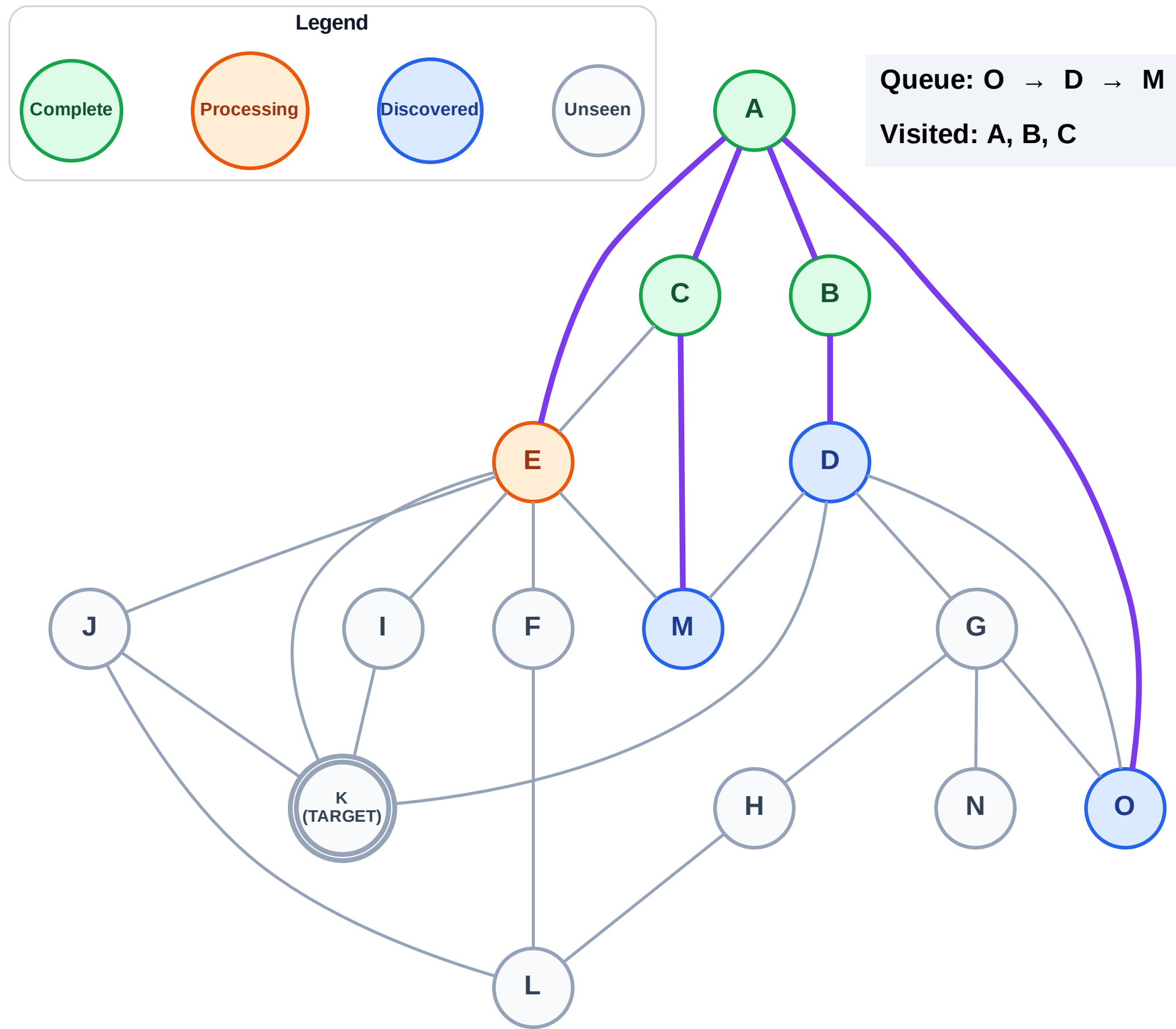


Visited: A, B, C



BFS Traversal

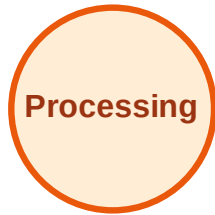
Step 8: Process node E



BFS Traversal

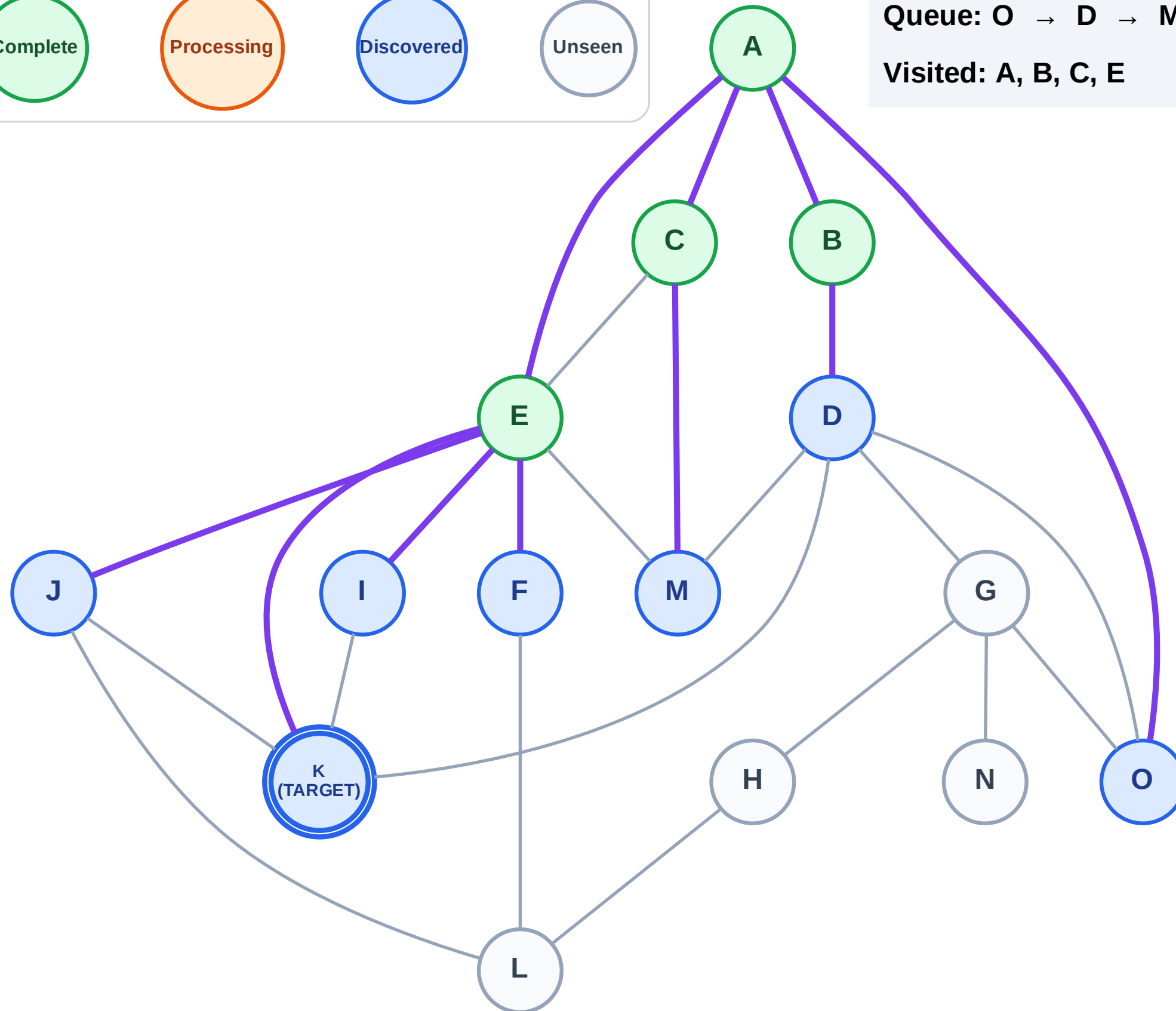
Step 9: Discover F, I, J, K from E

Legend



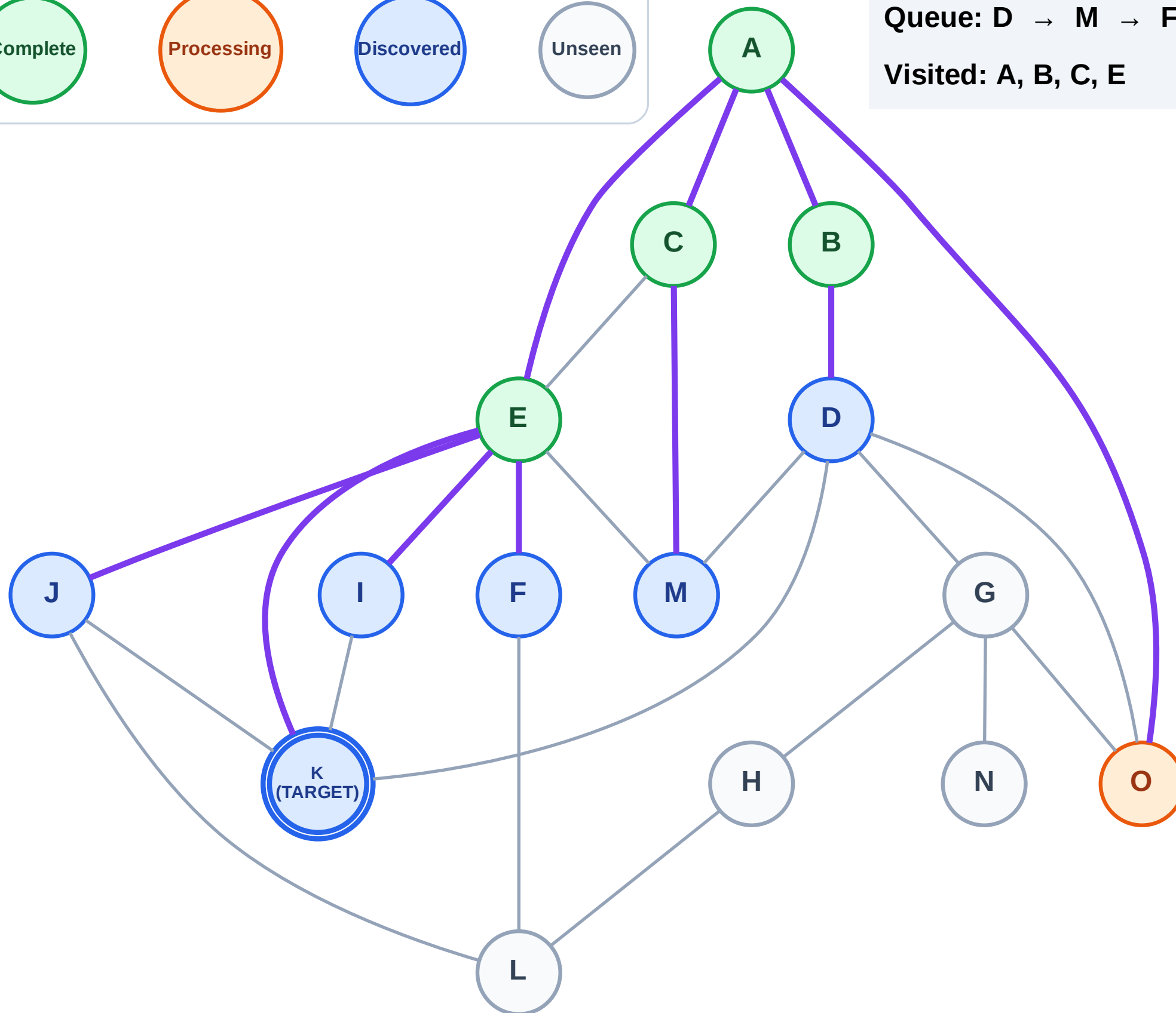
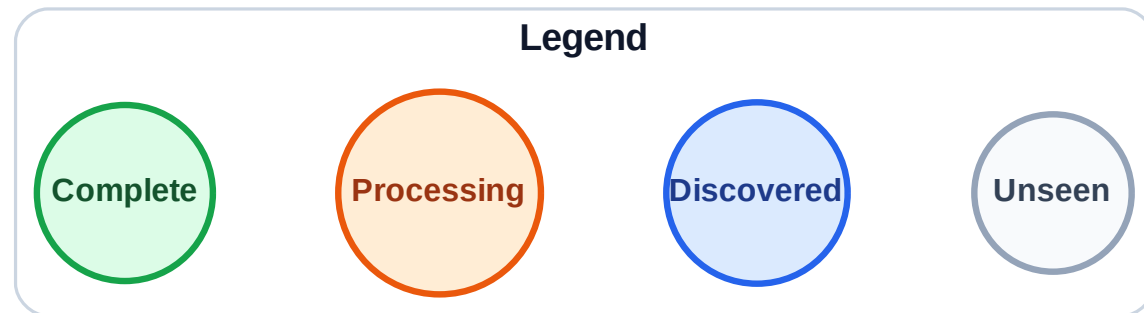
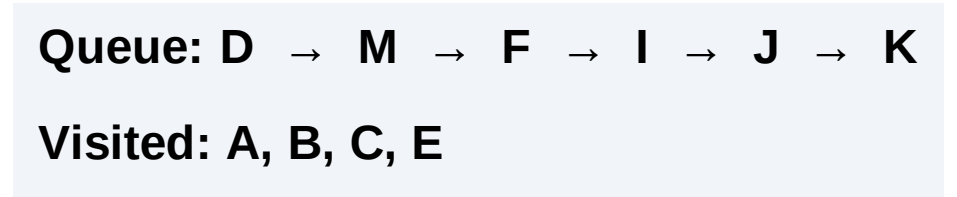
Queue: O → D → M → F → I → J → K

Visited: A, B, C, E



Step 10: Process node O

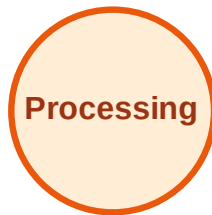
Step 10: Process node O



BFS Traversal

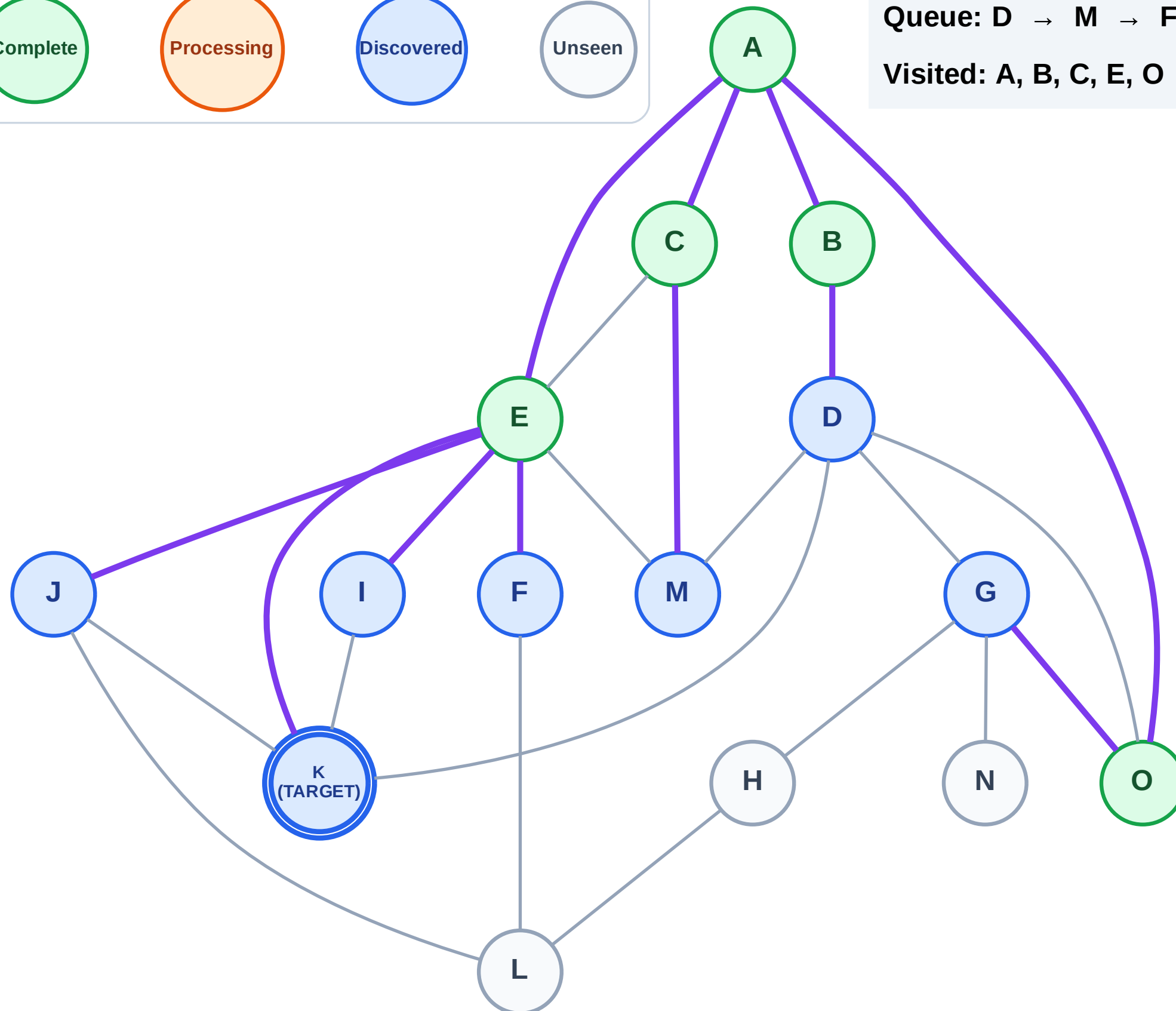
Step 11: Discover G from O

Legend



Queue: D → M → F → I → J → K → G

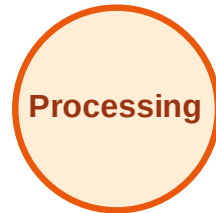
Visited: A, B, C, E, O



BFS Traversal

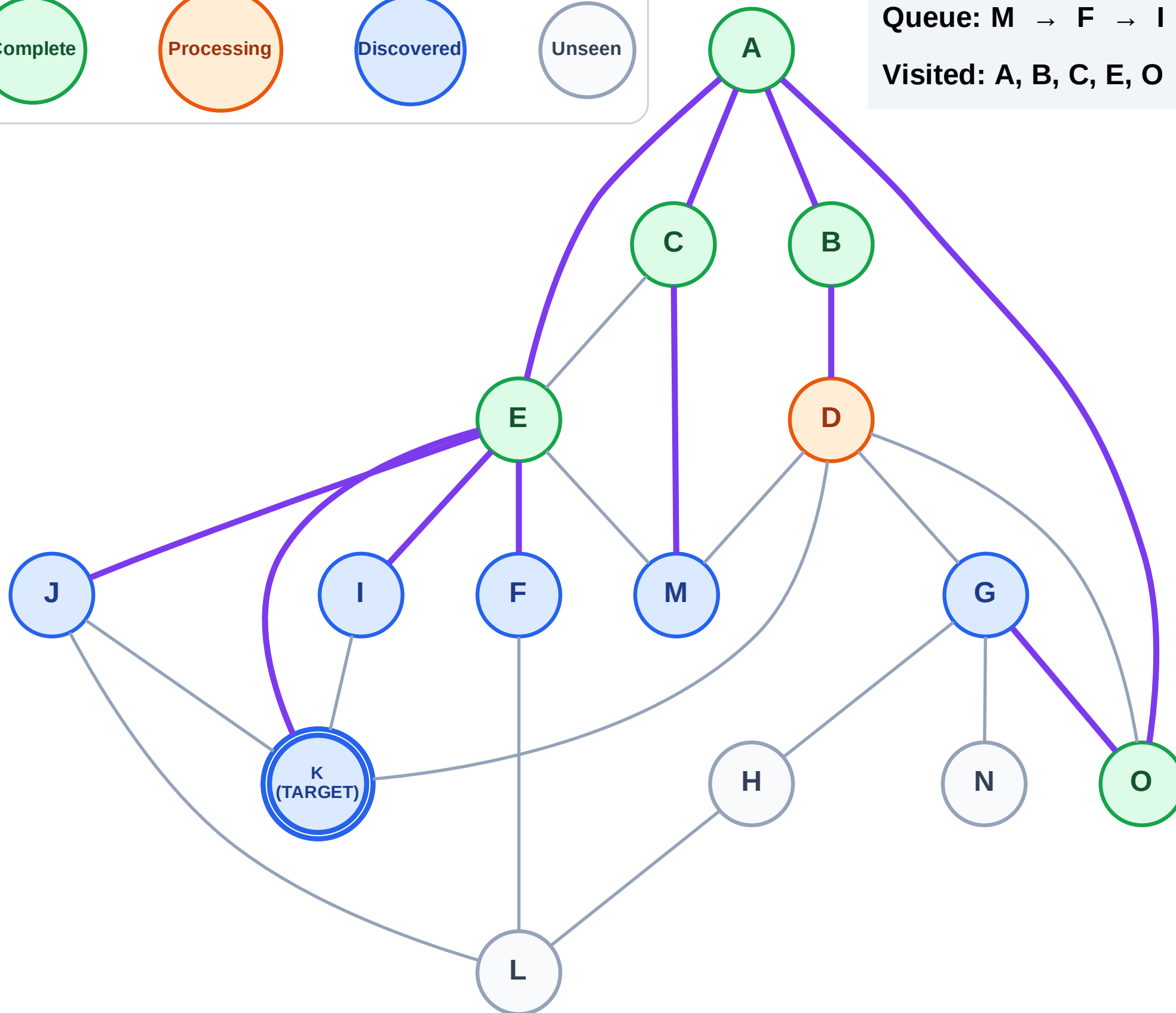
Step 12: Process node D

Legend



Queue: M → F → I → J → K → G

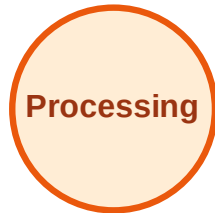
Visited: A, B, C, E, O



BFS Traversal

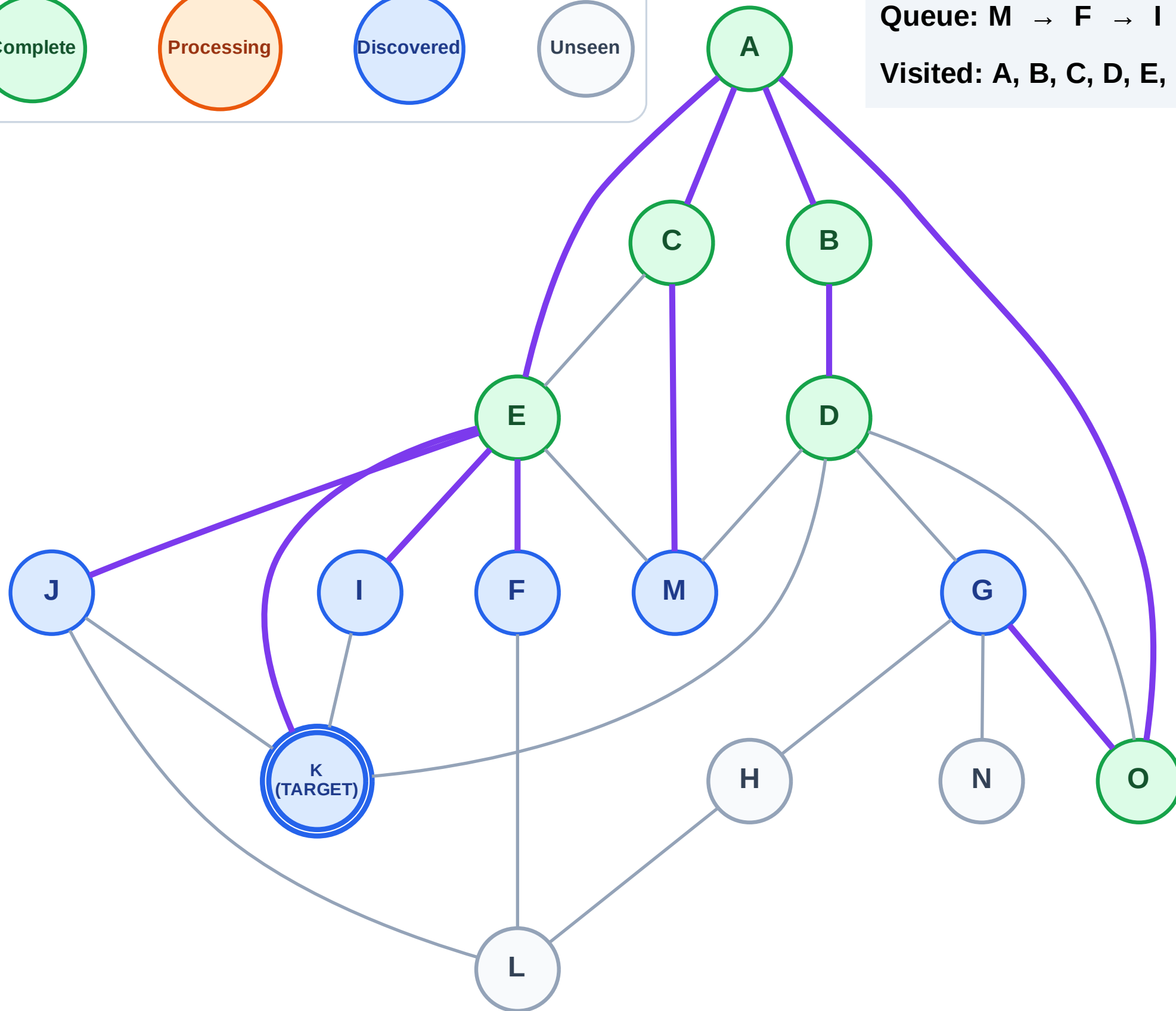
Step 13: No new neighbors from D

Legend



Queue: M → F → I → J → K → G

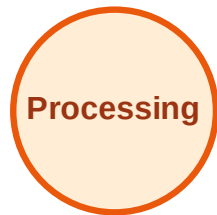
Visited: A, B, C, D, E, O



BFS Traversal

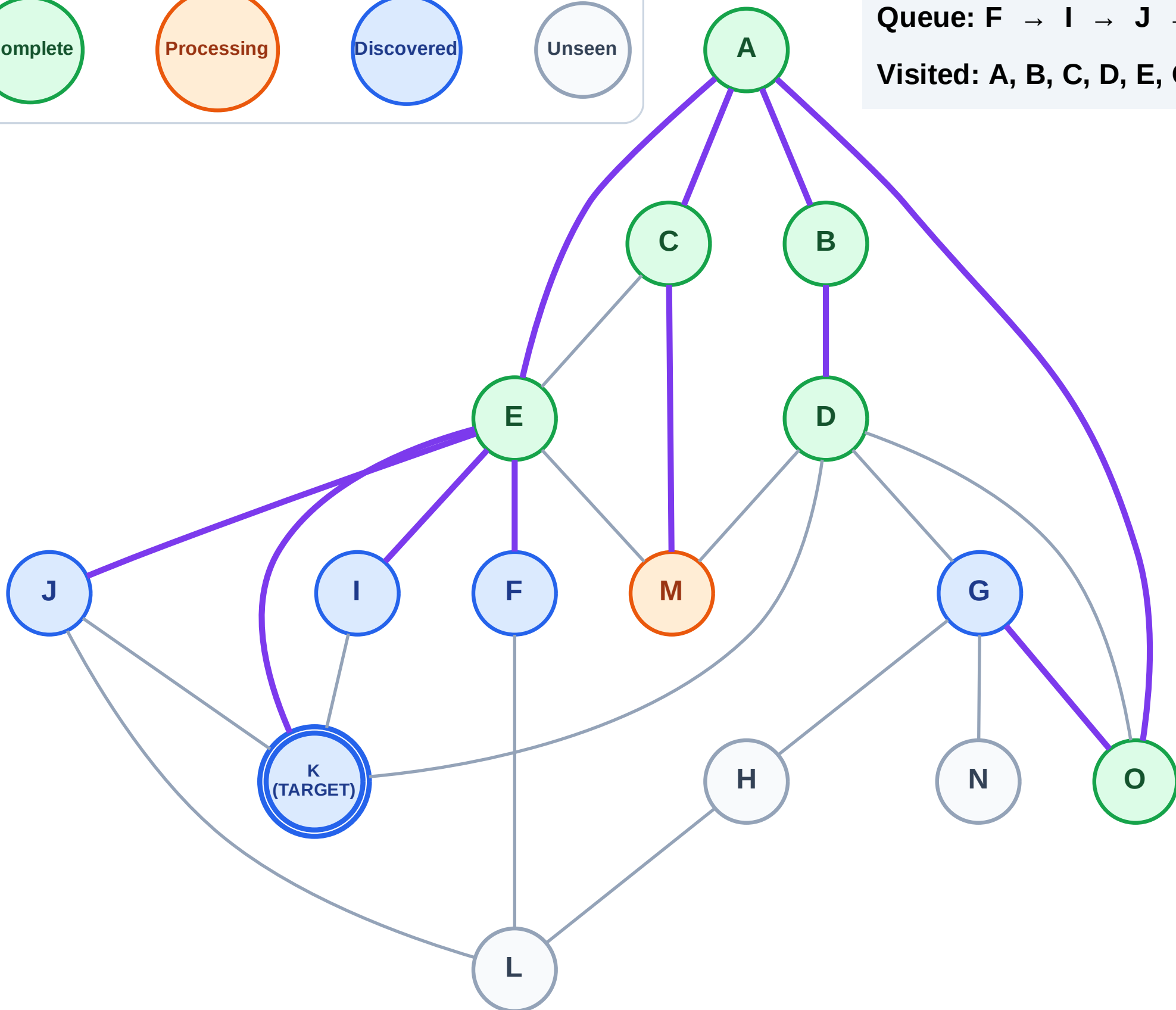
Step 14: Process node M

Legend



Queue: F → I → J → K → G

Visited: A, B, C, D, E, O



BFS Traversal

Step 15: No new neighbors from M

Legend

Complete

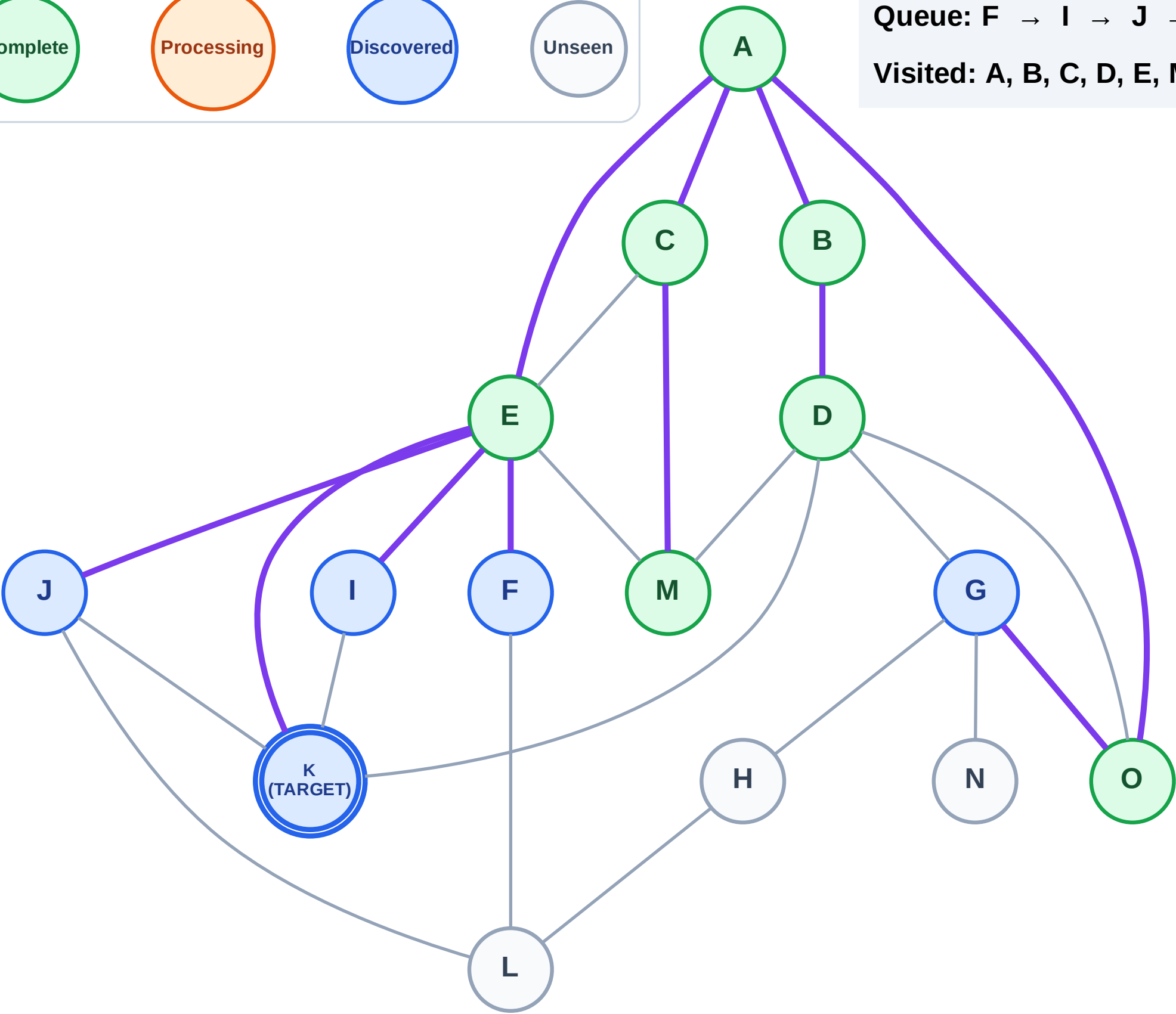
Processing

Discovered

Unseen

Queue: F → I → J → K → G

Visited: A, B, C, D, E, M, O



BFS Traversal

Step 16: Process node F

Legend

Complete

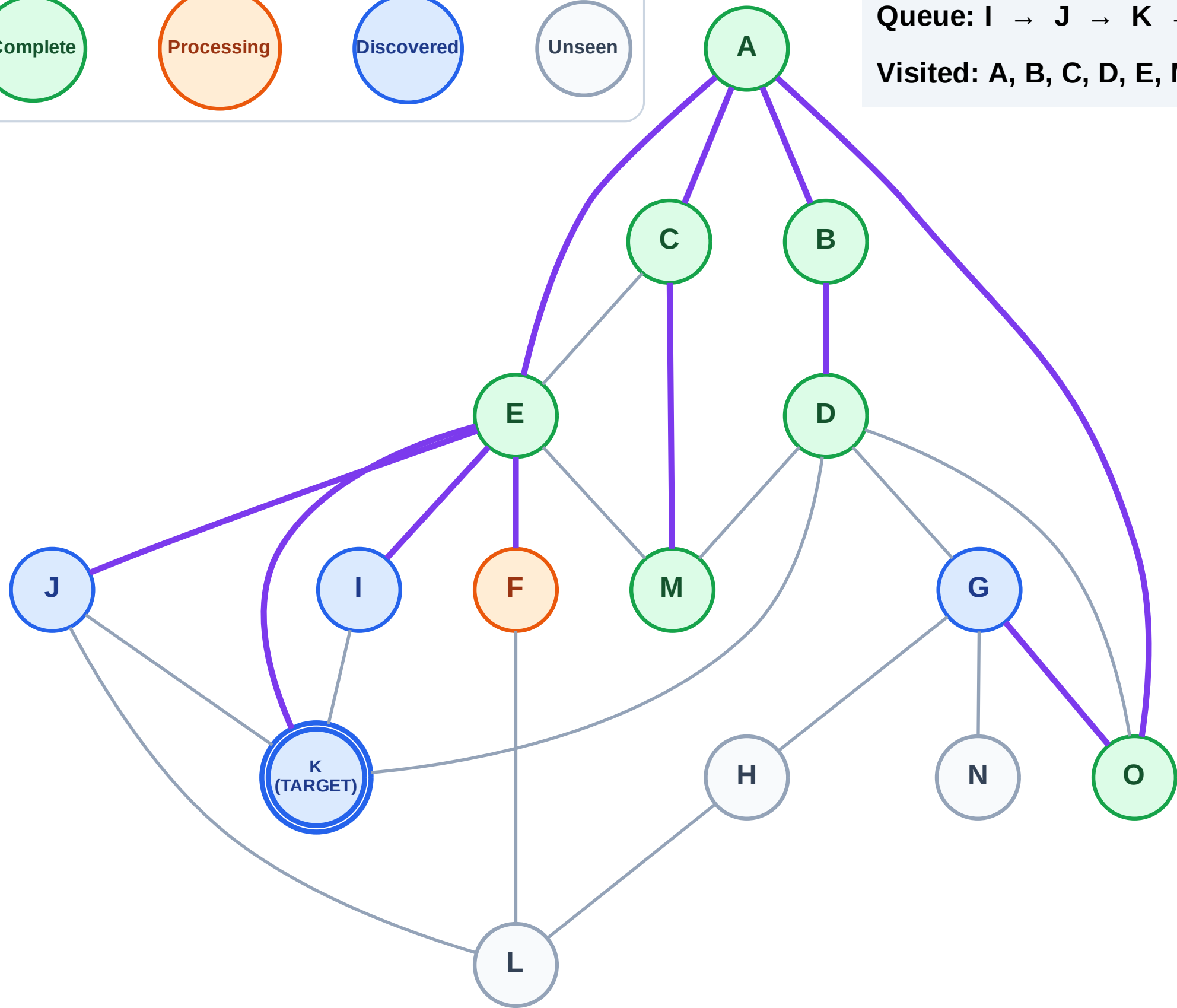
Processing

Discovered

Unseen

Queue: I → J → K → G

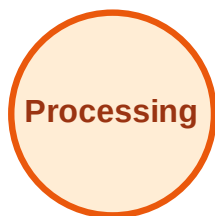
Visited: A, B, C, D, E, M, O



BFS Traversal

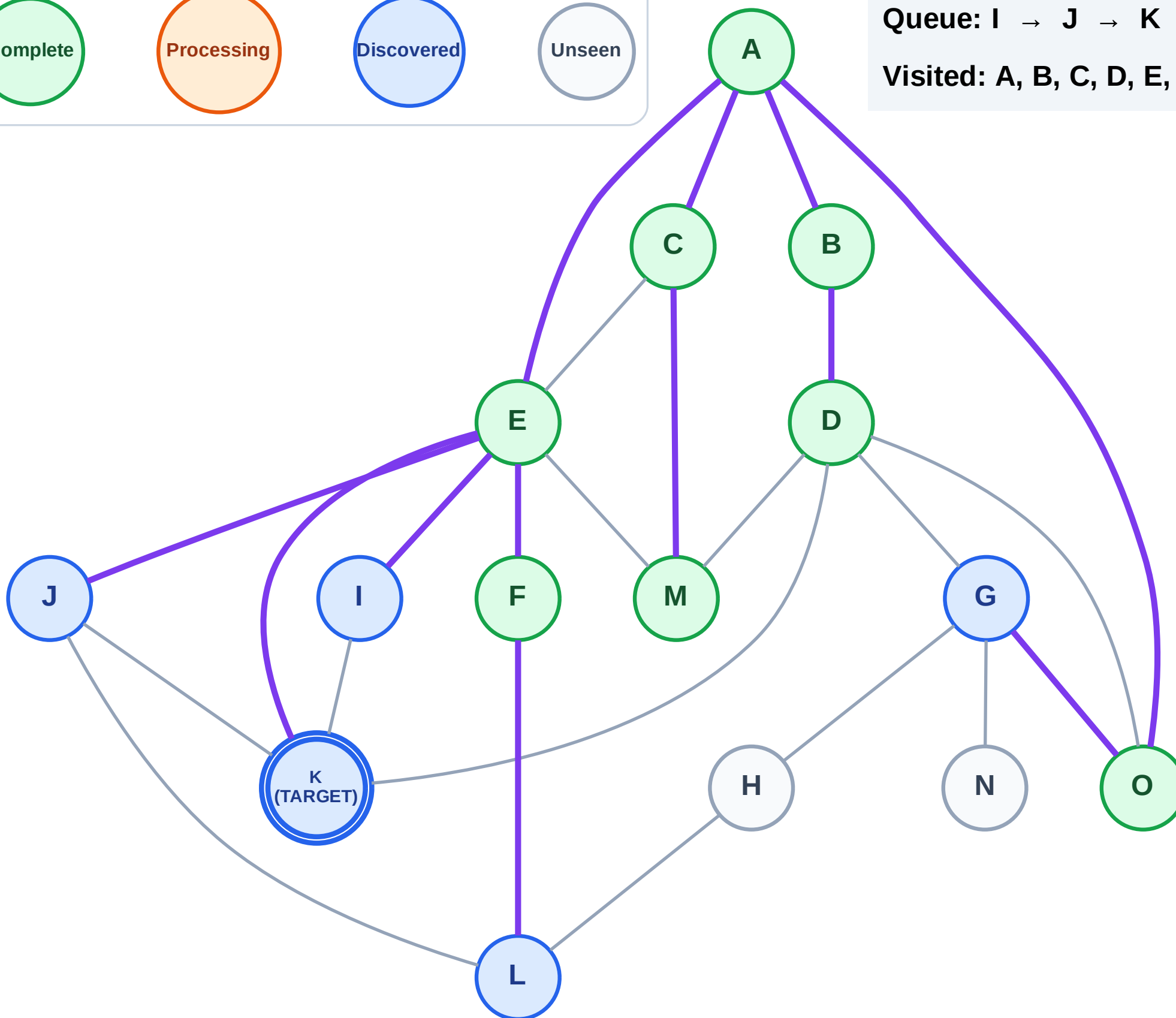
Step 17: Discover L from F

Legend



Queue: I → J → K → G → L

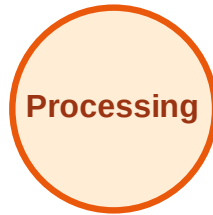
Visited: A, B, C, D, E, F, M, O



BFS Traversal

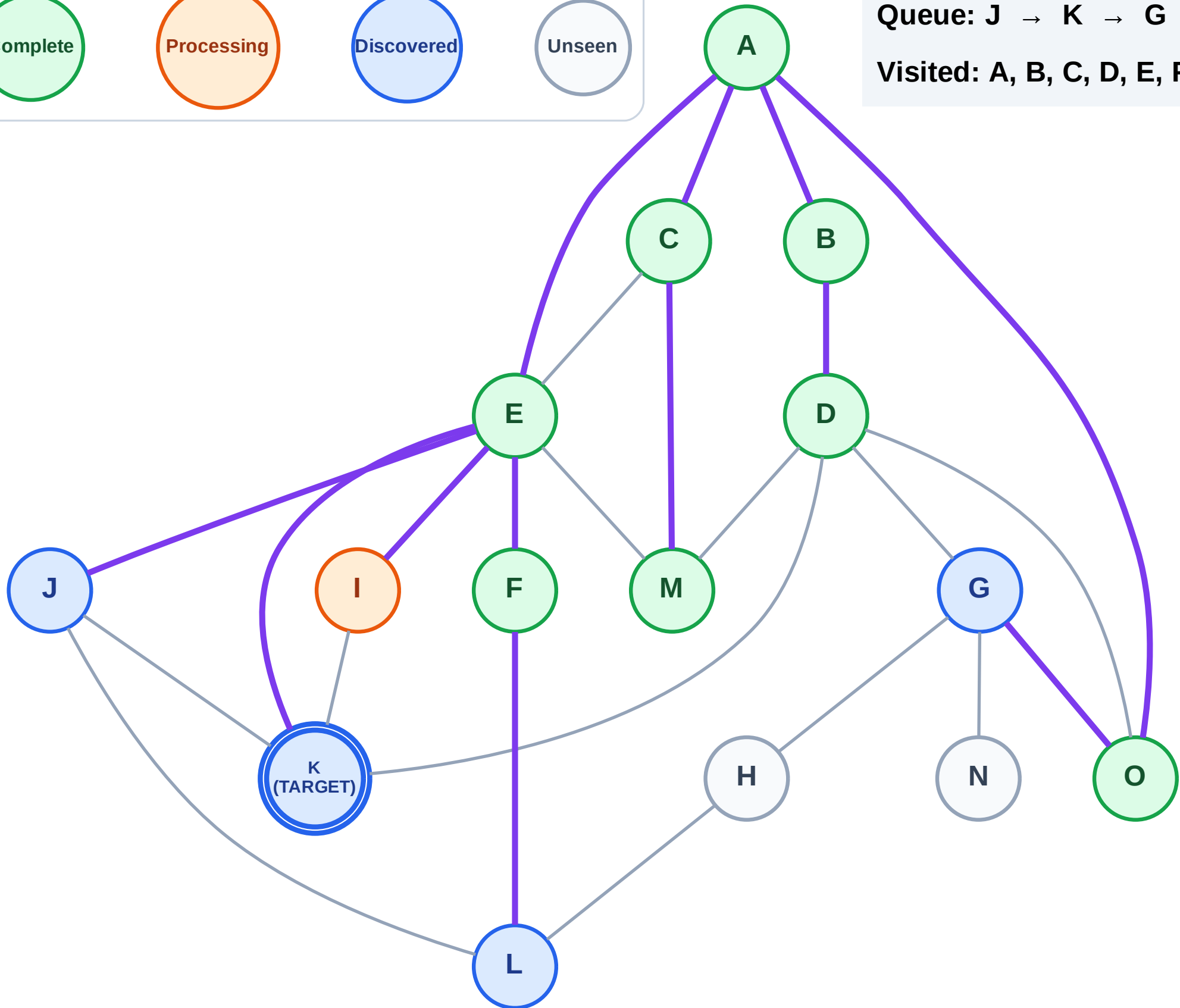
Step 18: Process node I

Legend



Queue: J → K → G → L

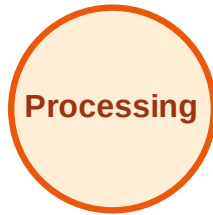
Visited: A, B, C, D, E, F, M, O



BFS Traversal

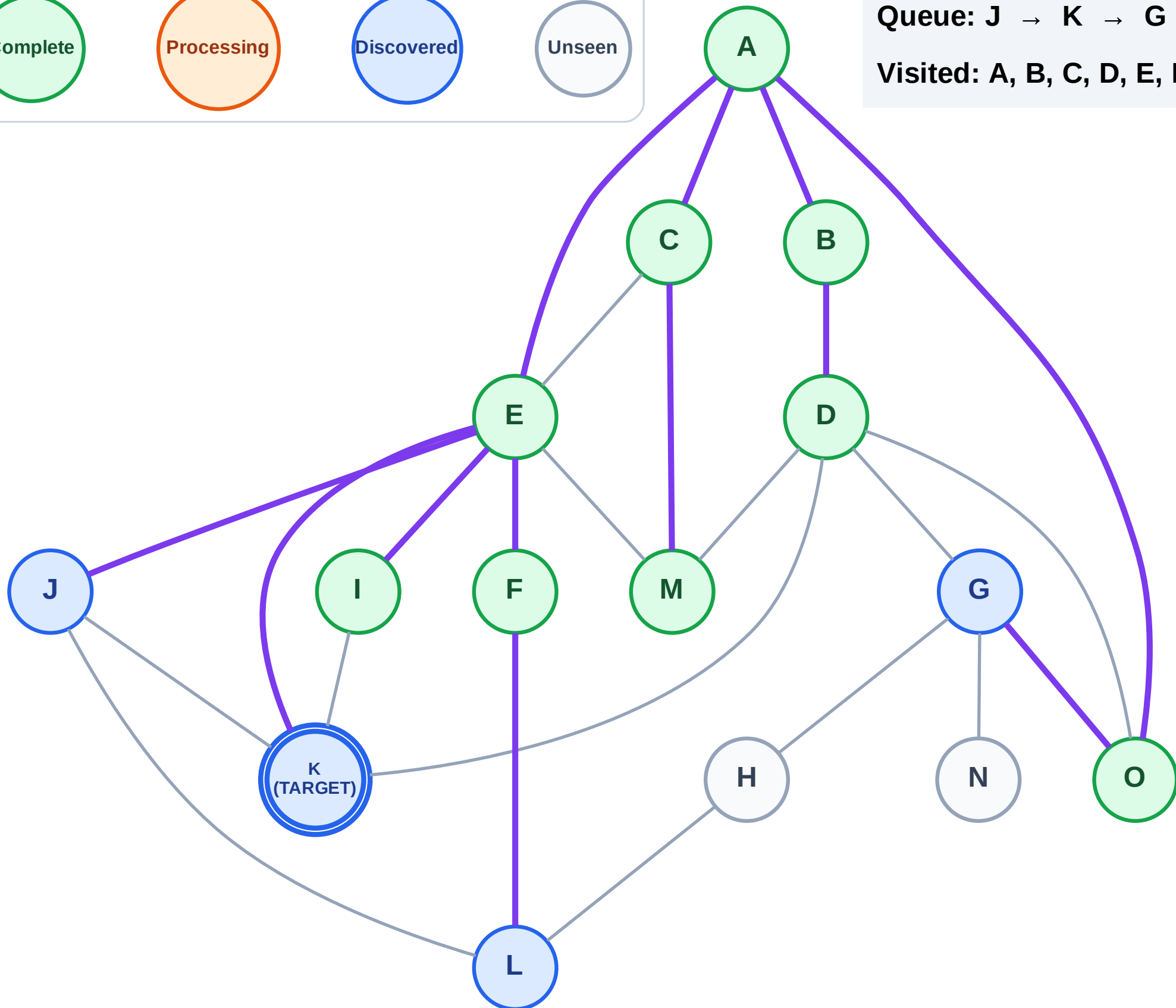
Step 19: No new neighbors from I

Legend



Queue: J → K → G → L

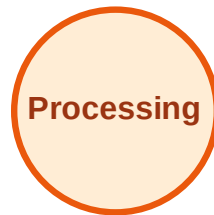
Visited: A, B, C, D, E, F, I, M, O



BFS Traversal

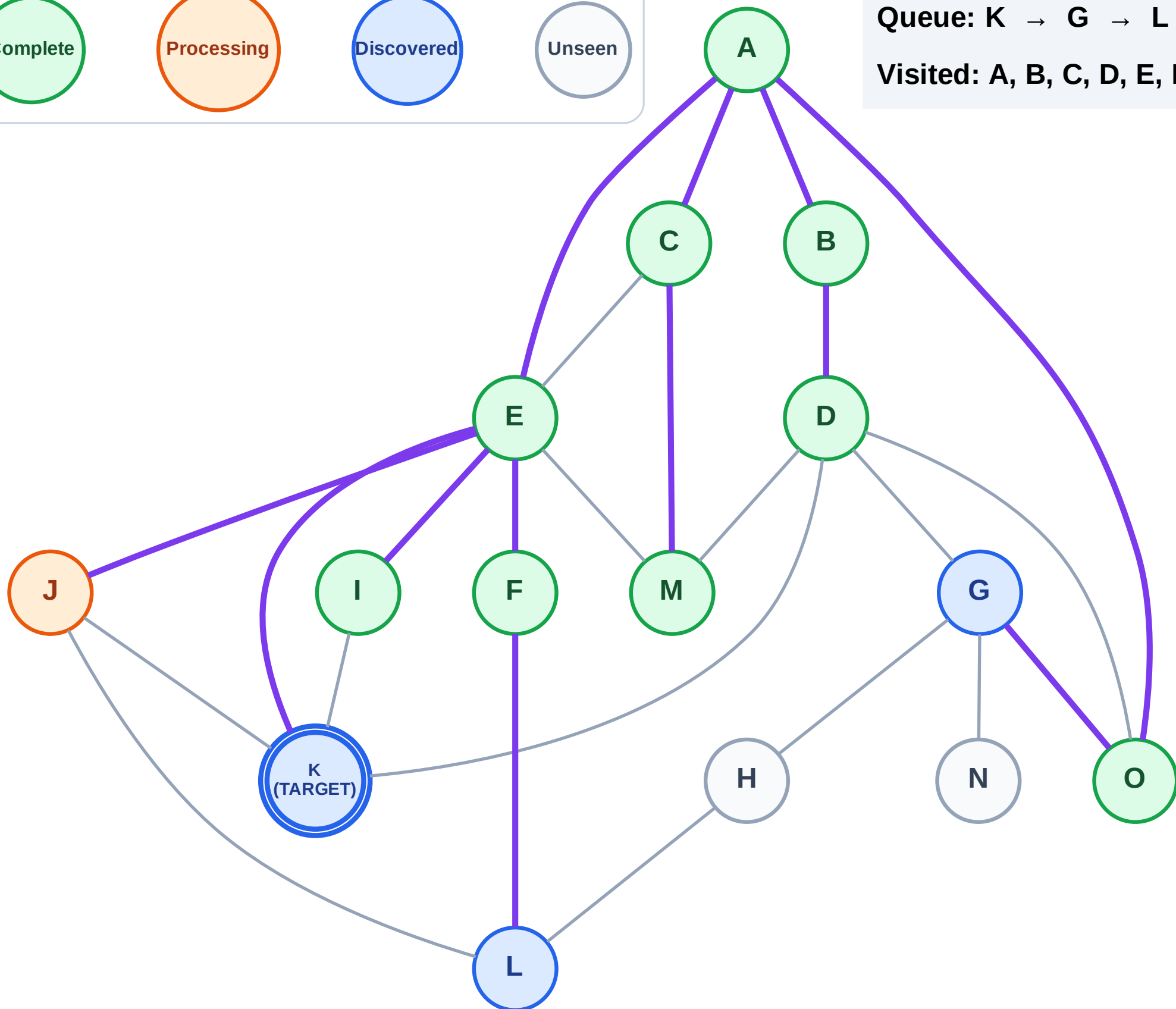
Step 20: Process node J

Legend



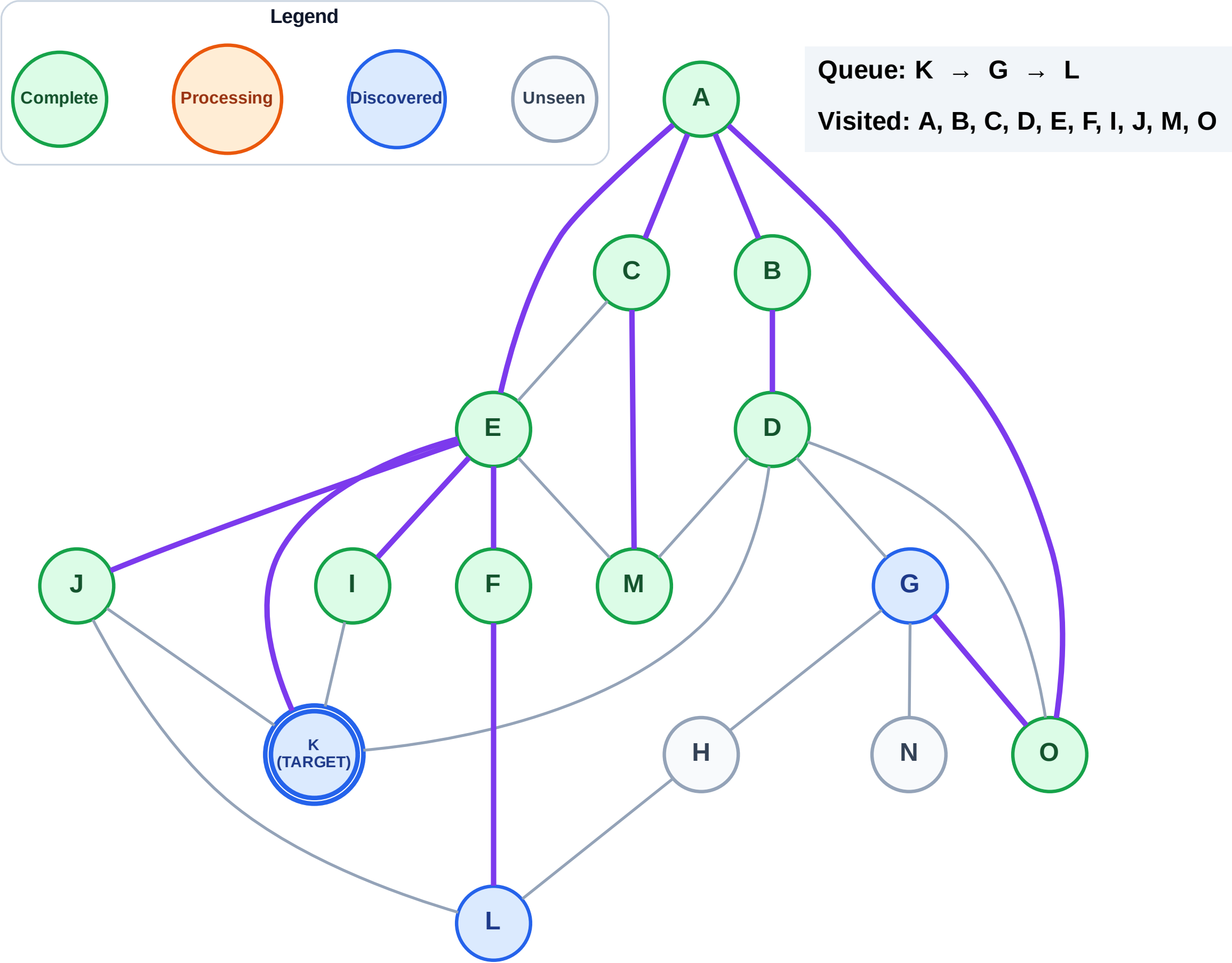
Queue: K → G → L

Visited: A, B, C, D, E, F, I, M, O



BFS Traversal

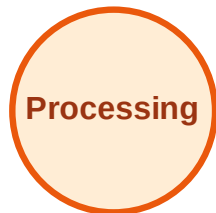
Step 21: No new neighbors from J



BFS Traversal

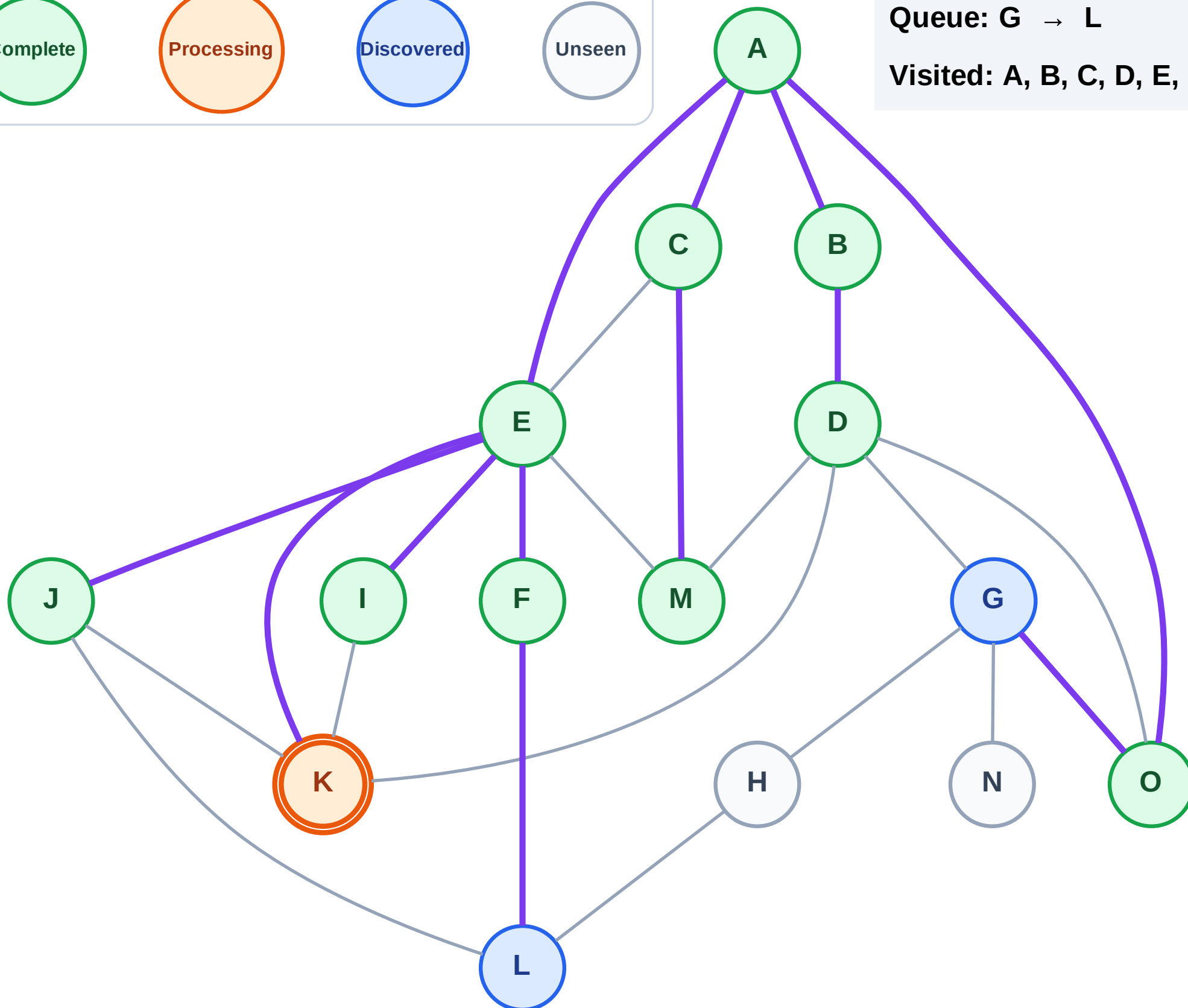
Step 22: Process node K

Legend



Queue: G → L

Visited: A, B, C, D, E, F, I, J, M, O



BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend

Complete

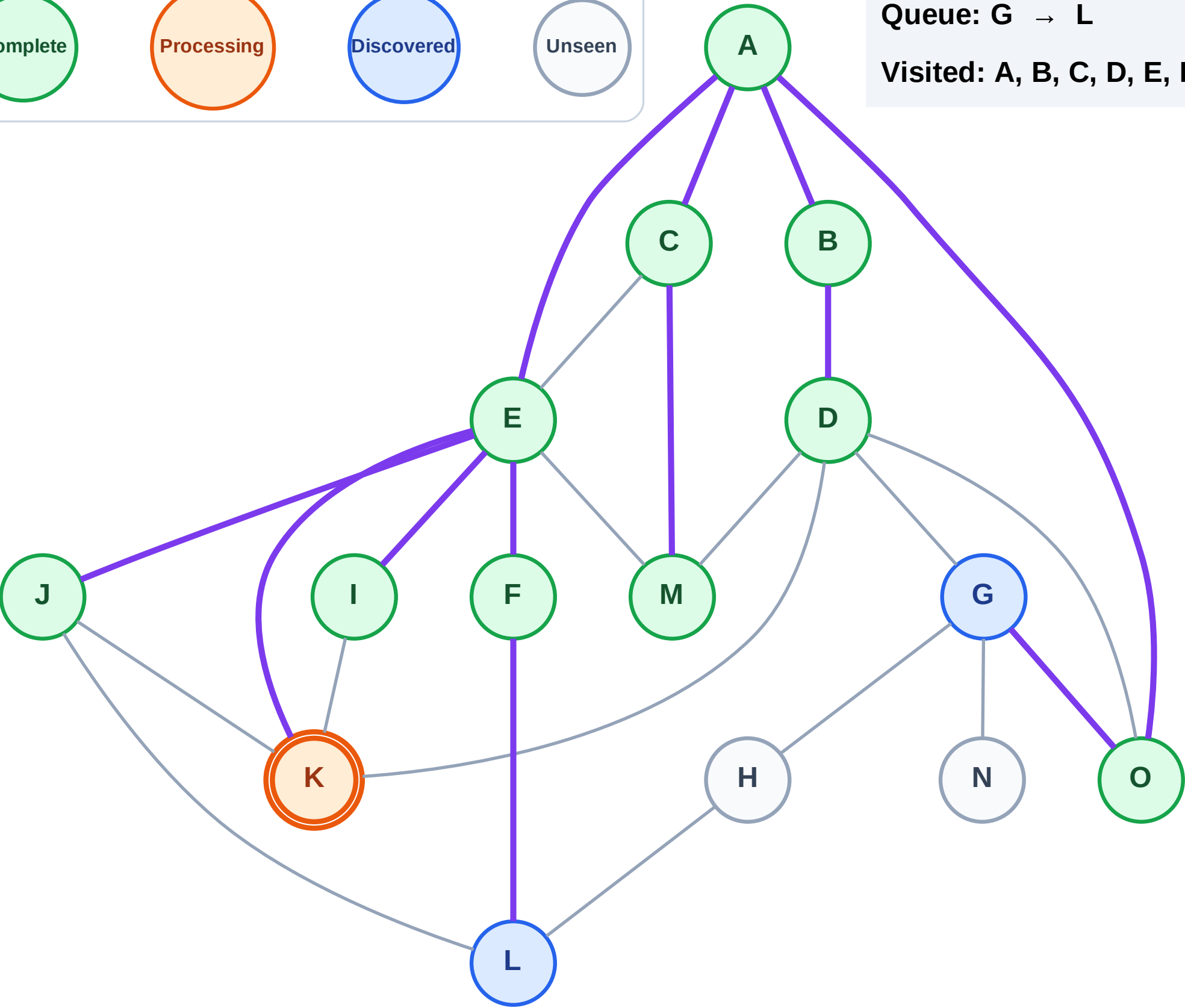
Processing

Discovered

Unseen

Queue: G → L

Visited: A, B, C, D, E, F, I, J, K, M, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

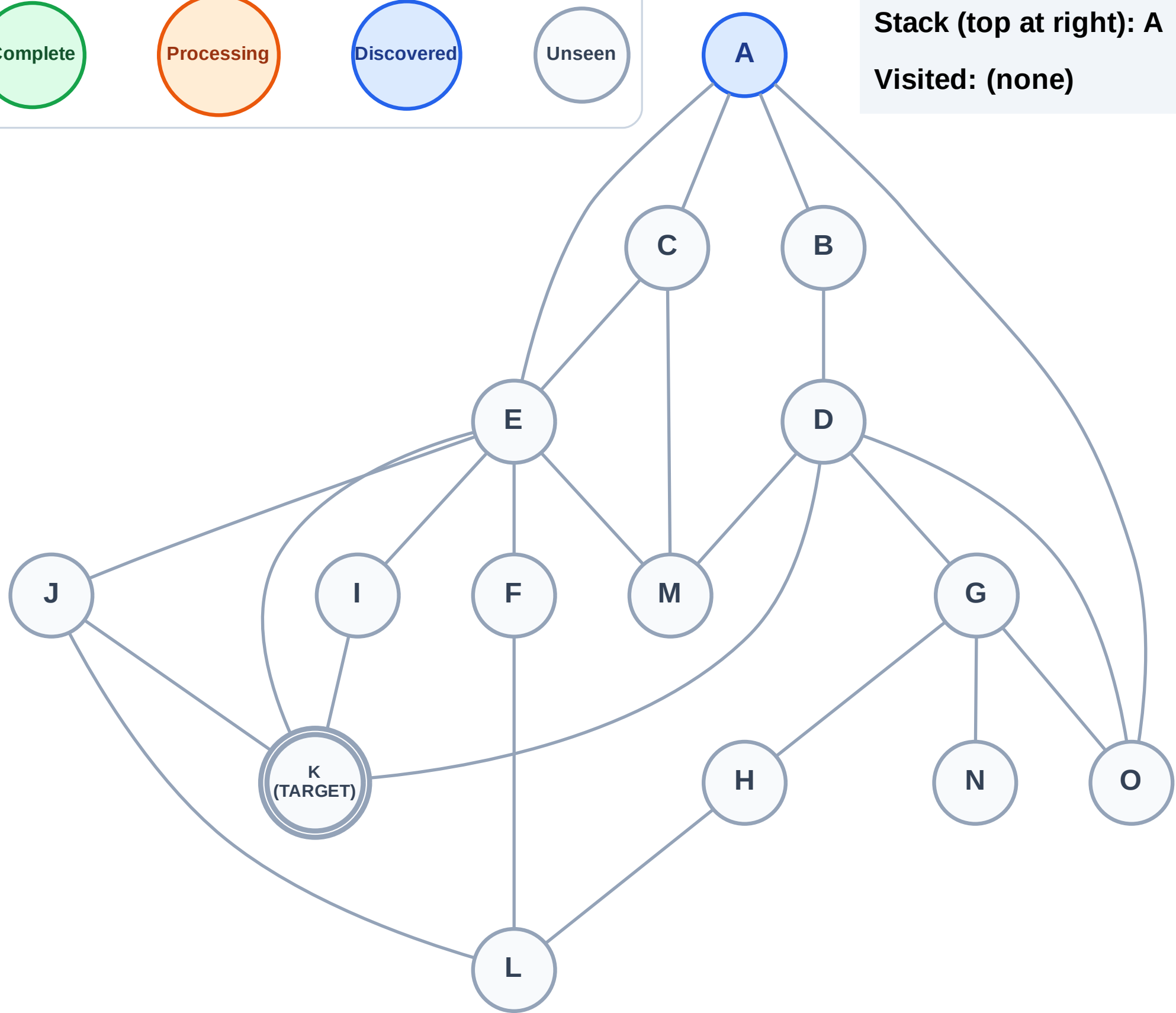
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

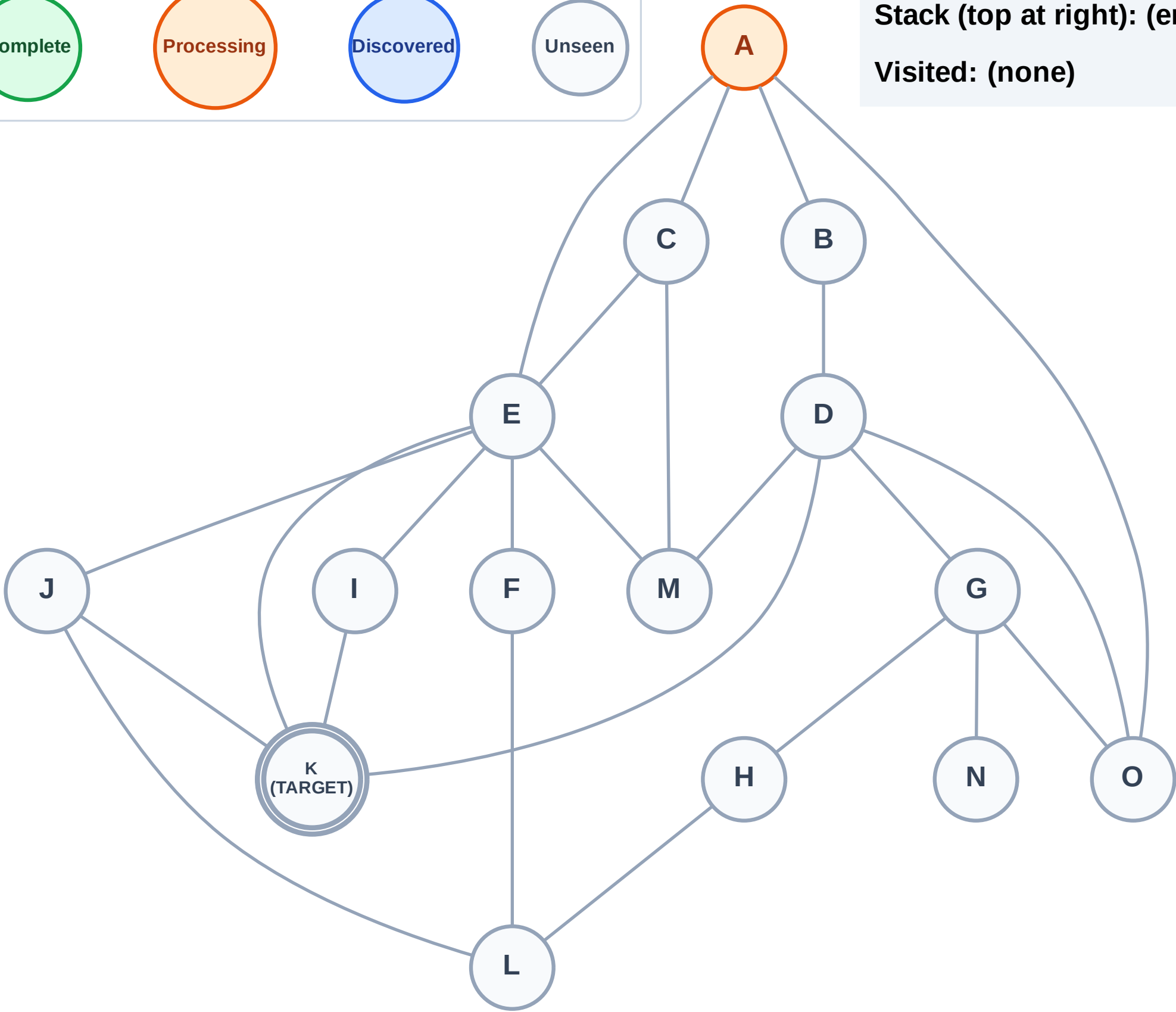
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover O, E, C, B from A

Legend

Complete

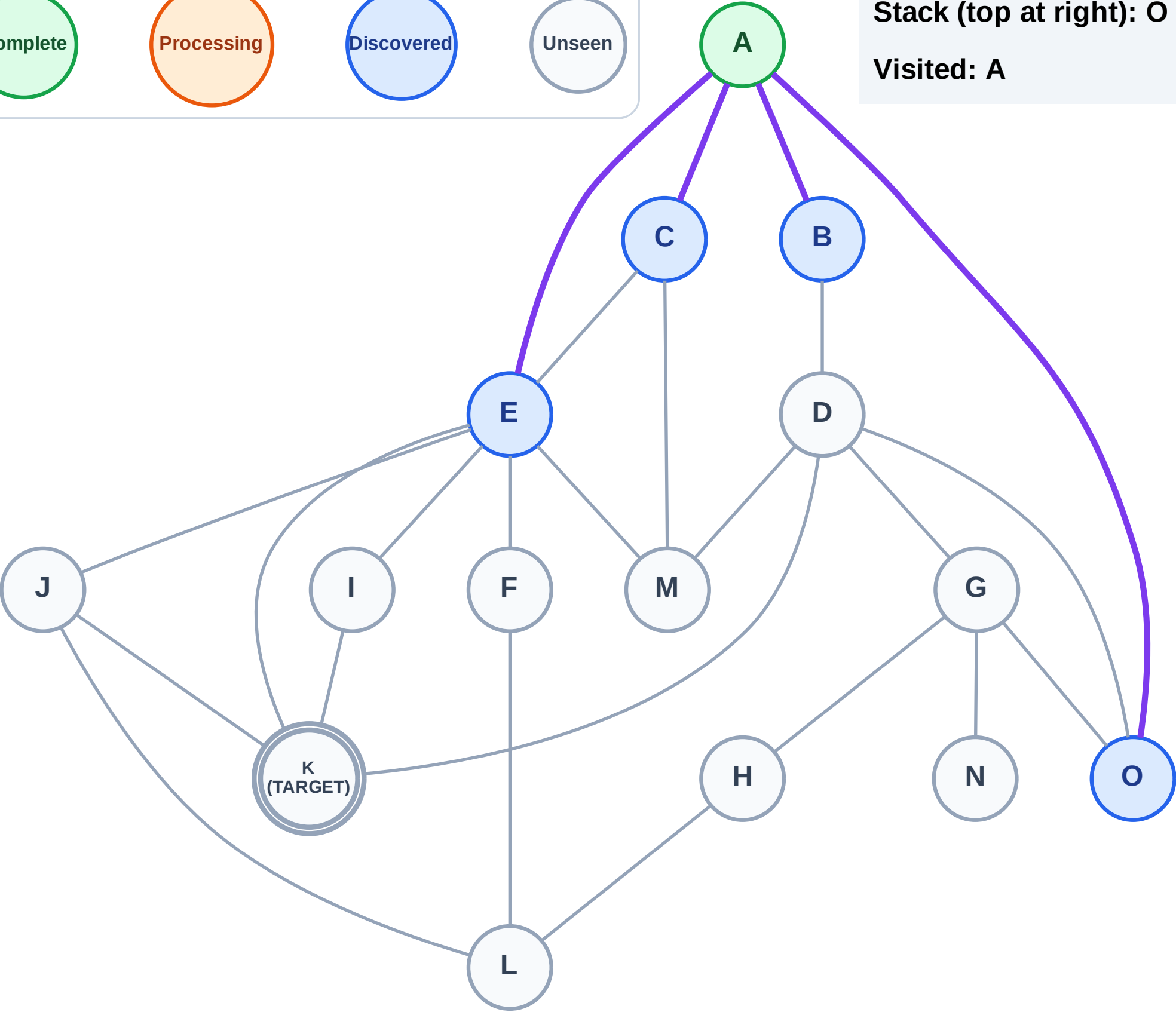
Processing

Discovered

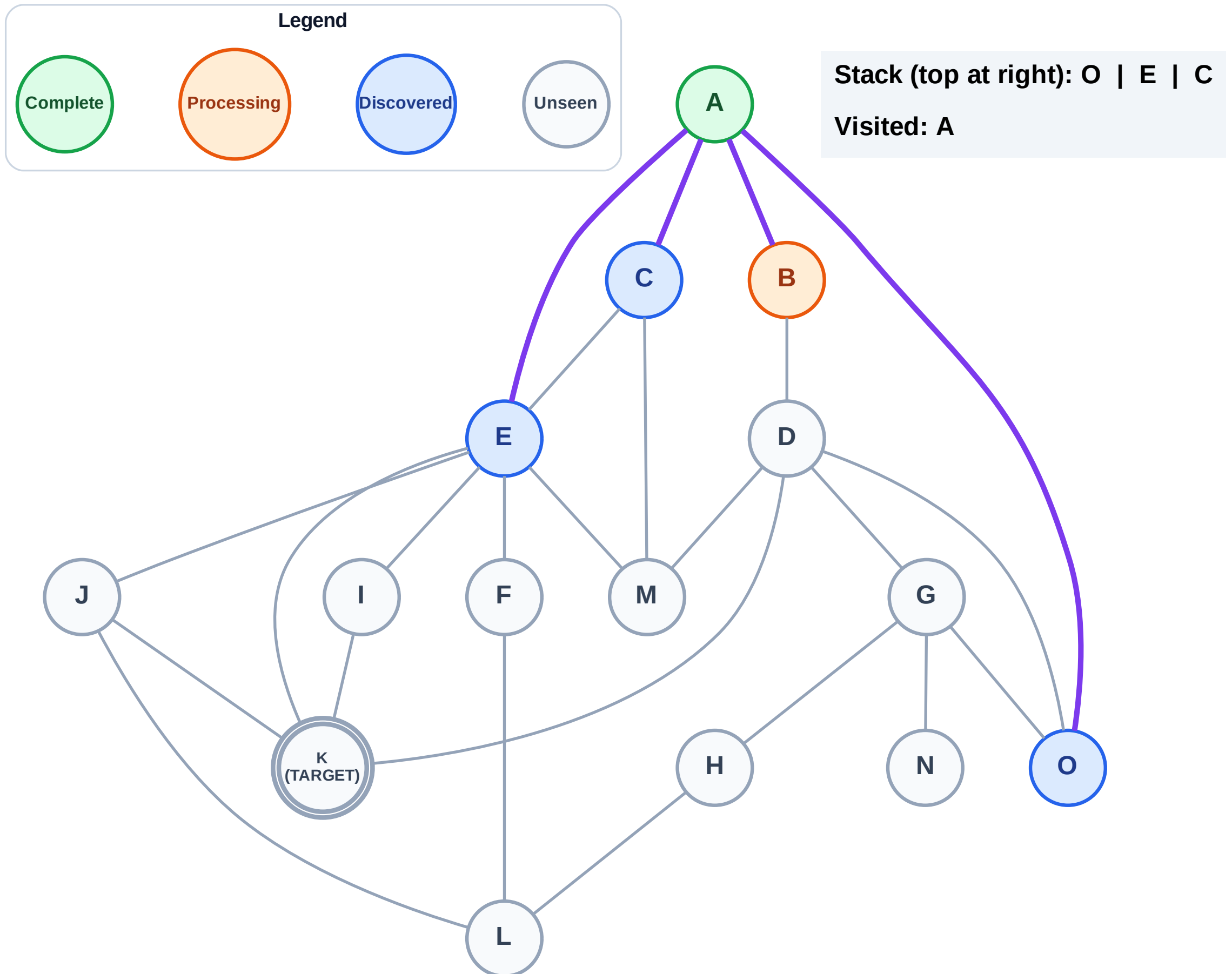
Unseen

Stack (top at right): O | E | C | B

Visited: A



Step 4: Process node B

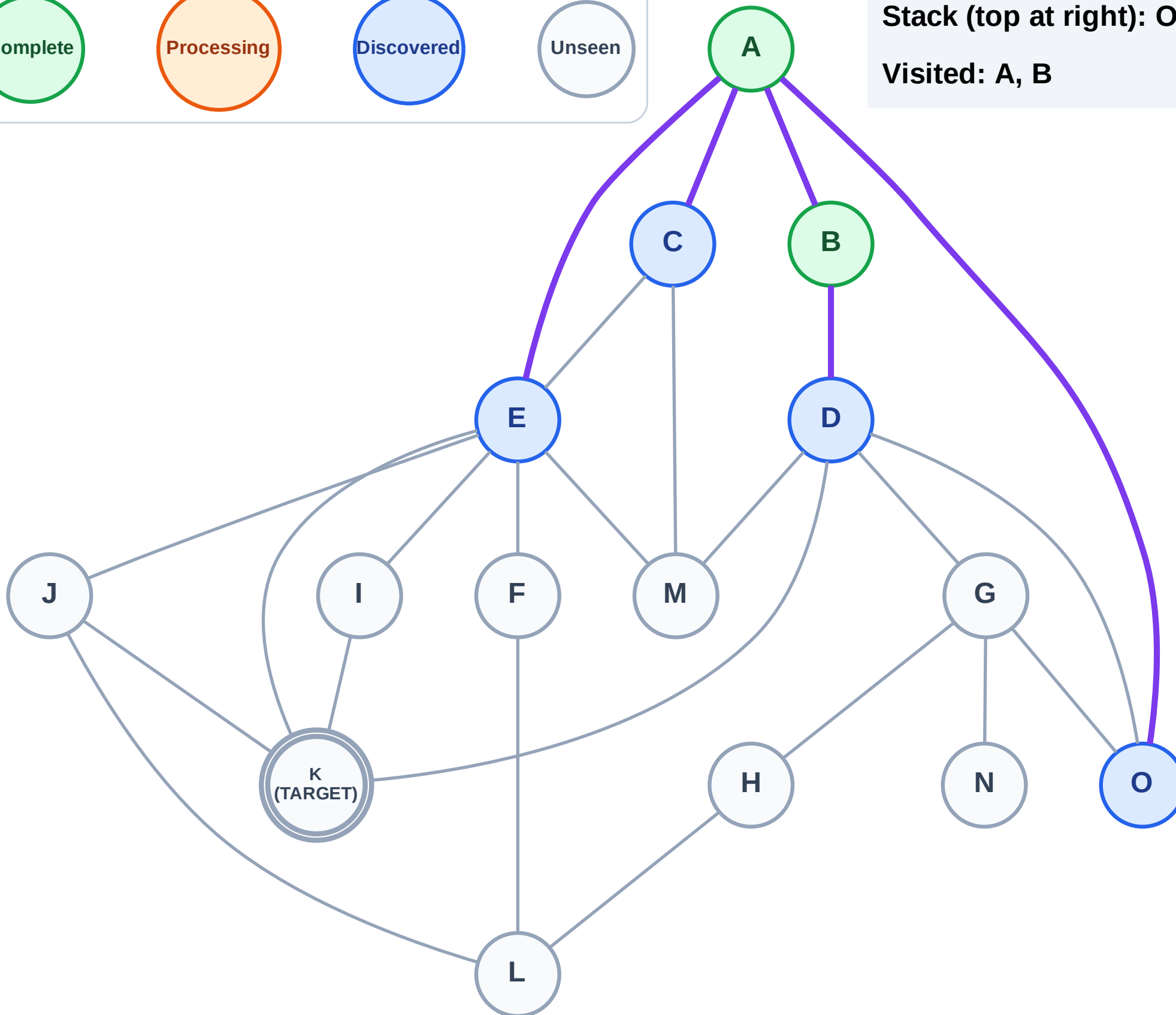


Step 5: Discover D from B

Legend



Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

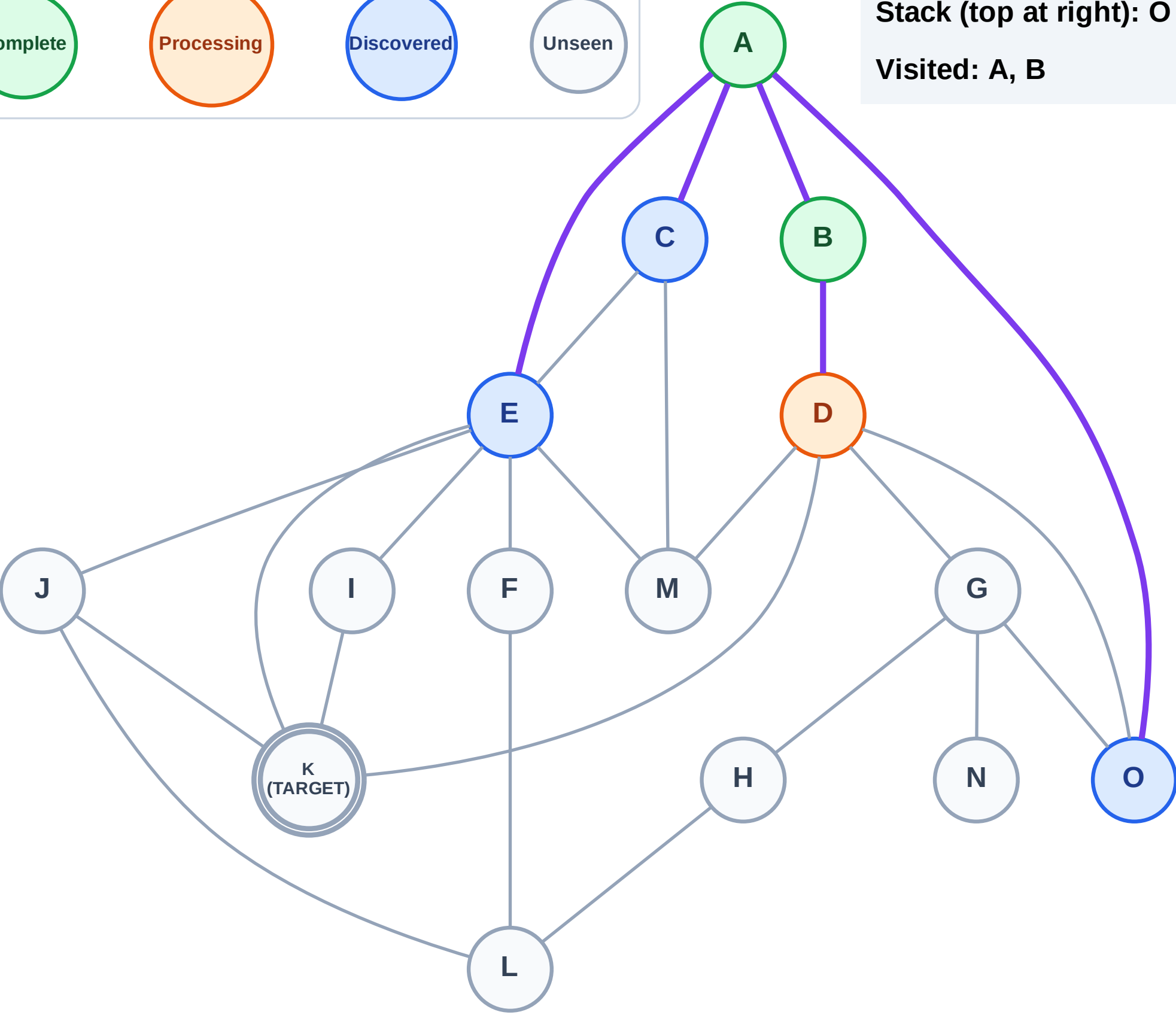
Complete

Processing

Discovered

Unseen

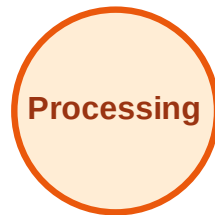
Stack (top at right): O | E | C
Visited: A, B



DFS Traversal

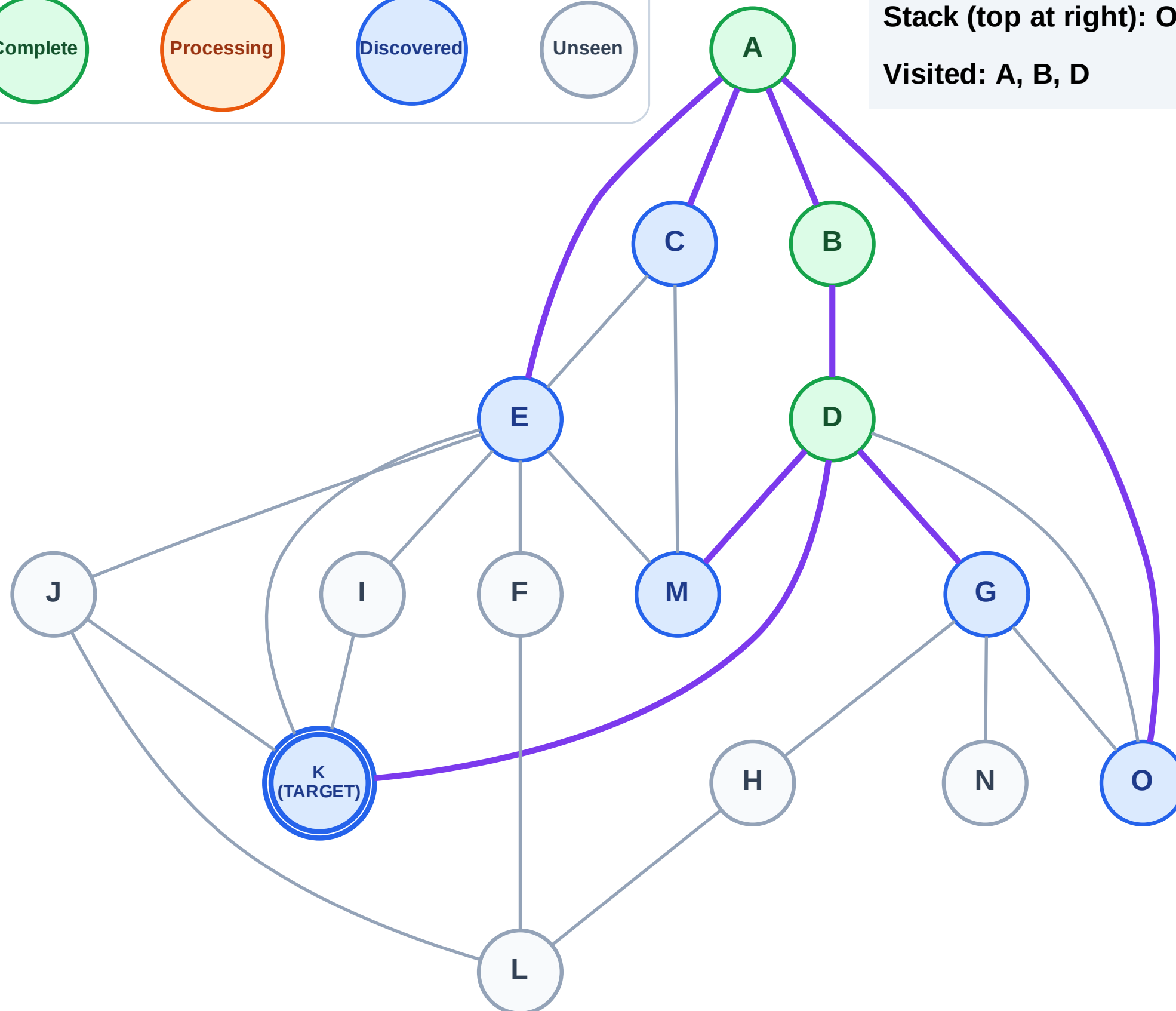
Step 7: Discover M, K, G from D

Legend



Stack (top at right): O | E | C | M | K | G

Visited: A, B, D



DFS Traversal

Step 8: Process node G

Legend

Complete

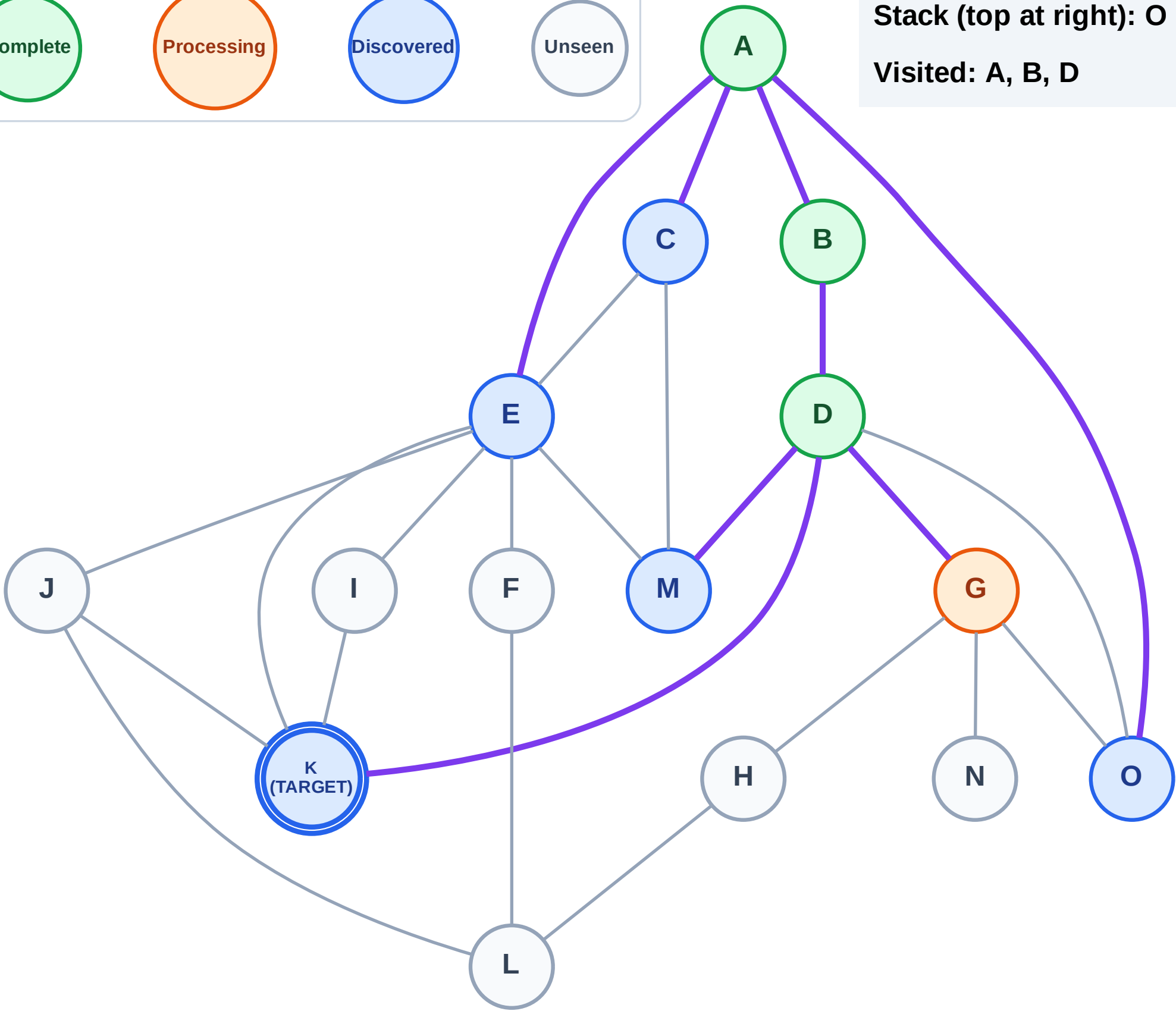
Processing

Discovered

Unseen

Stack (top at right): O | E | C | M | K

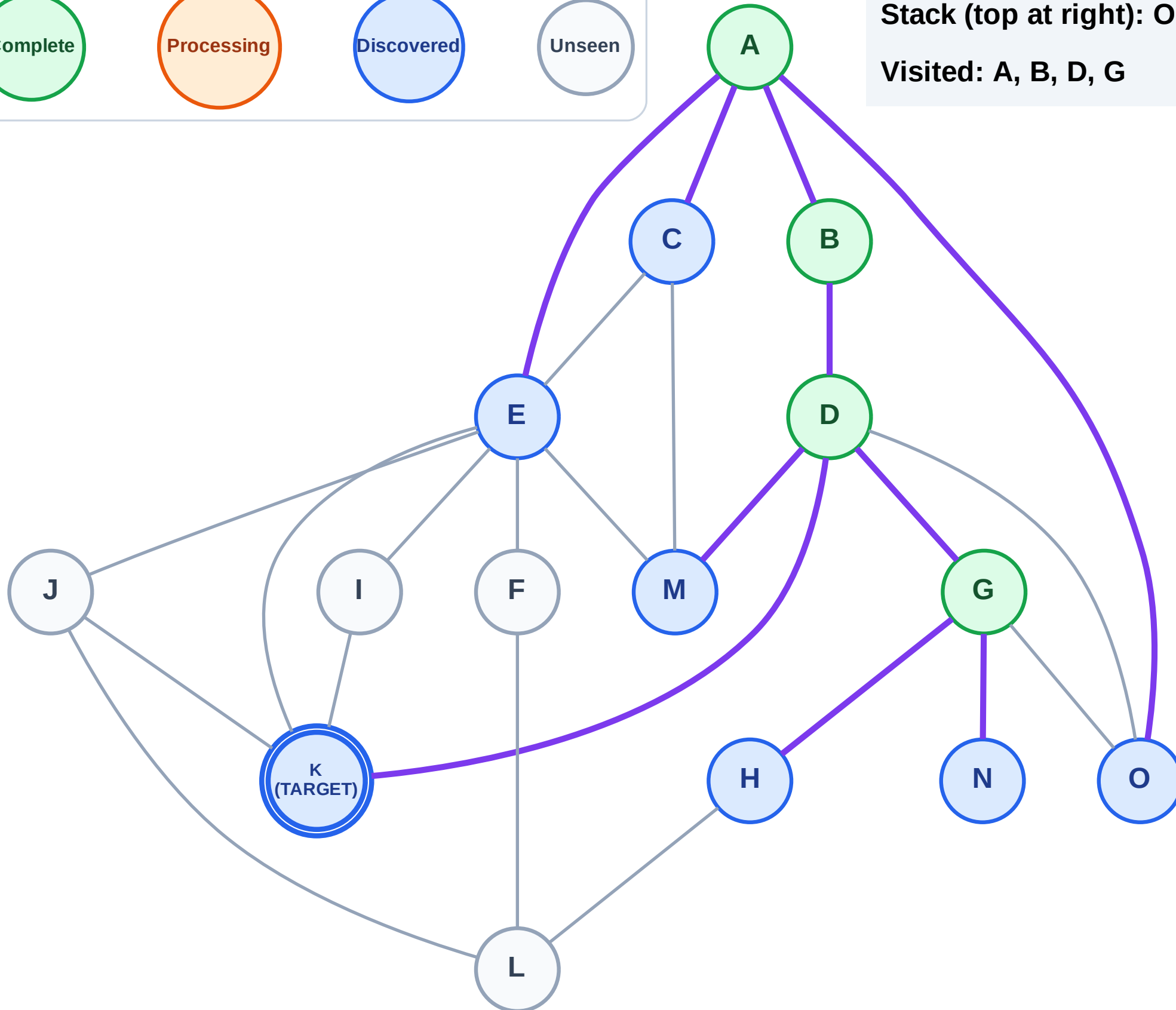
Visited: A, B, D



Step 9: Discover N, H from G

Stack (top at right): O | E | C | M | K | N | H

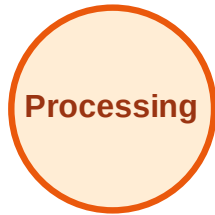
Visited: A, B, D, G



DFS Traversal

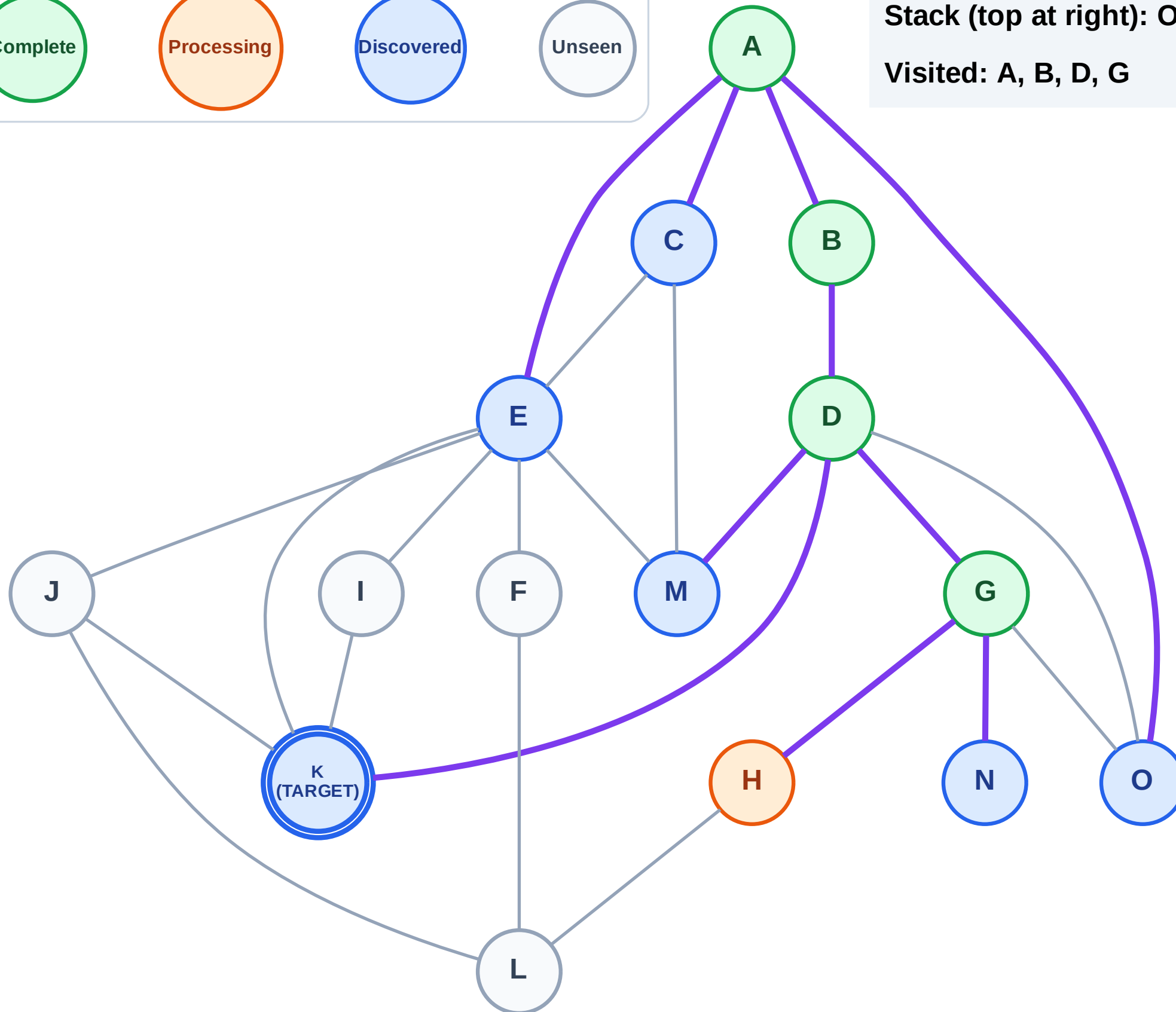
Step 10: Process node H

Legend



Stack (top at right): O | E | C | M | K | N

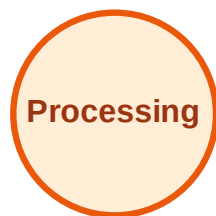
Visited: A, B, D, G



DFS Traversal

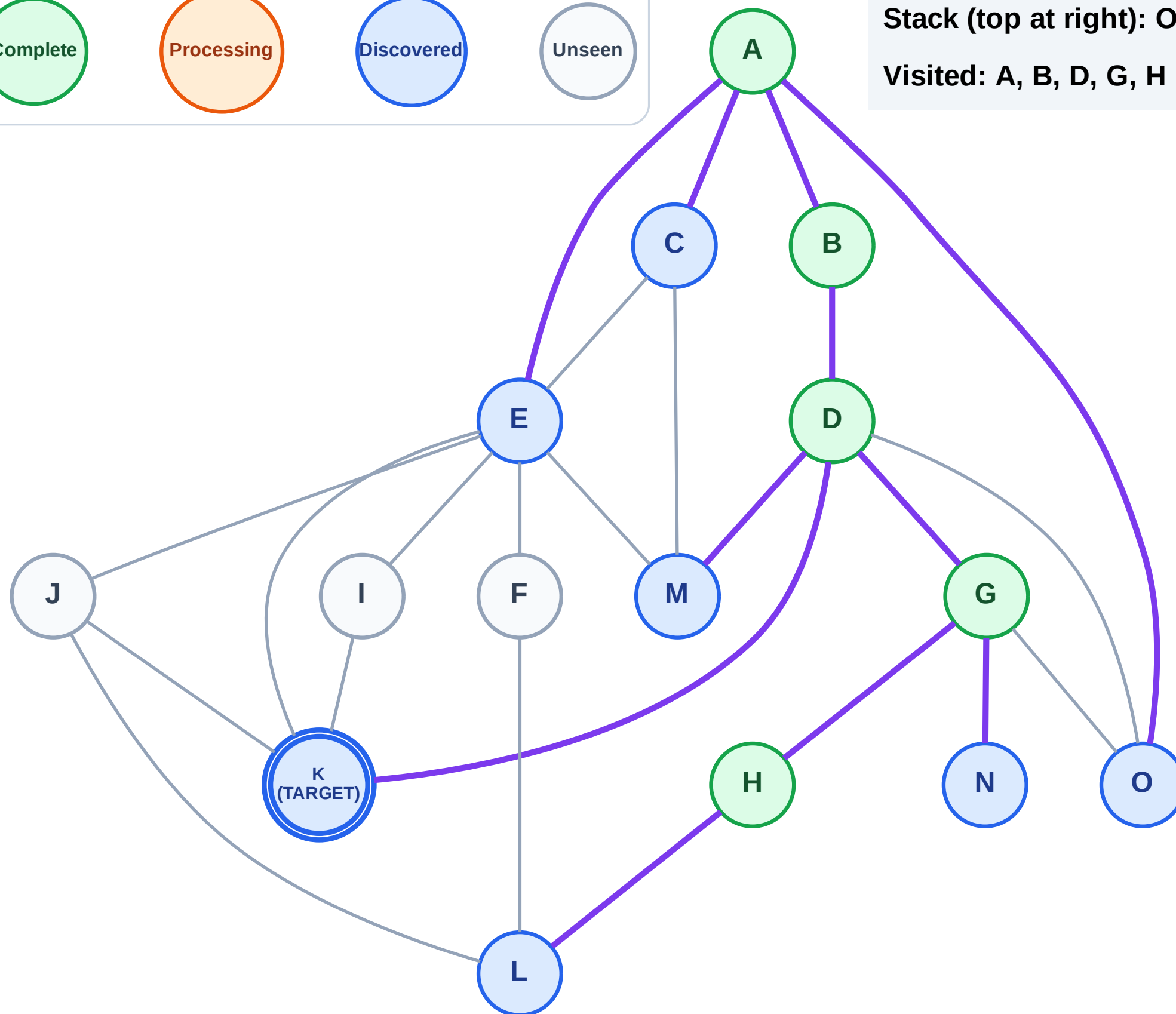
Step 11: Discover L from H

Legend



Stack (top at right): O | E | C | M | K | N | L

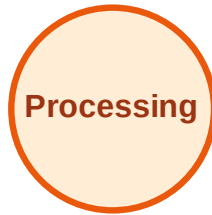
Visited: A, B, D, G, H



DFS Traversal

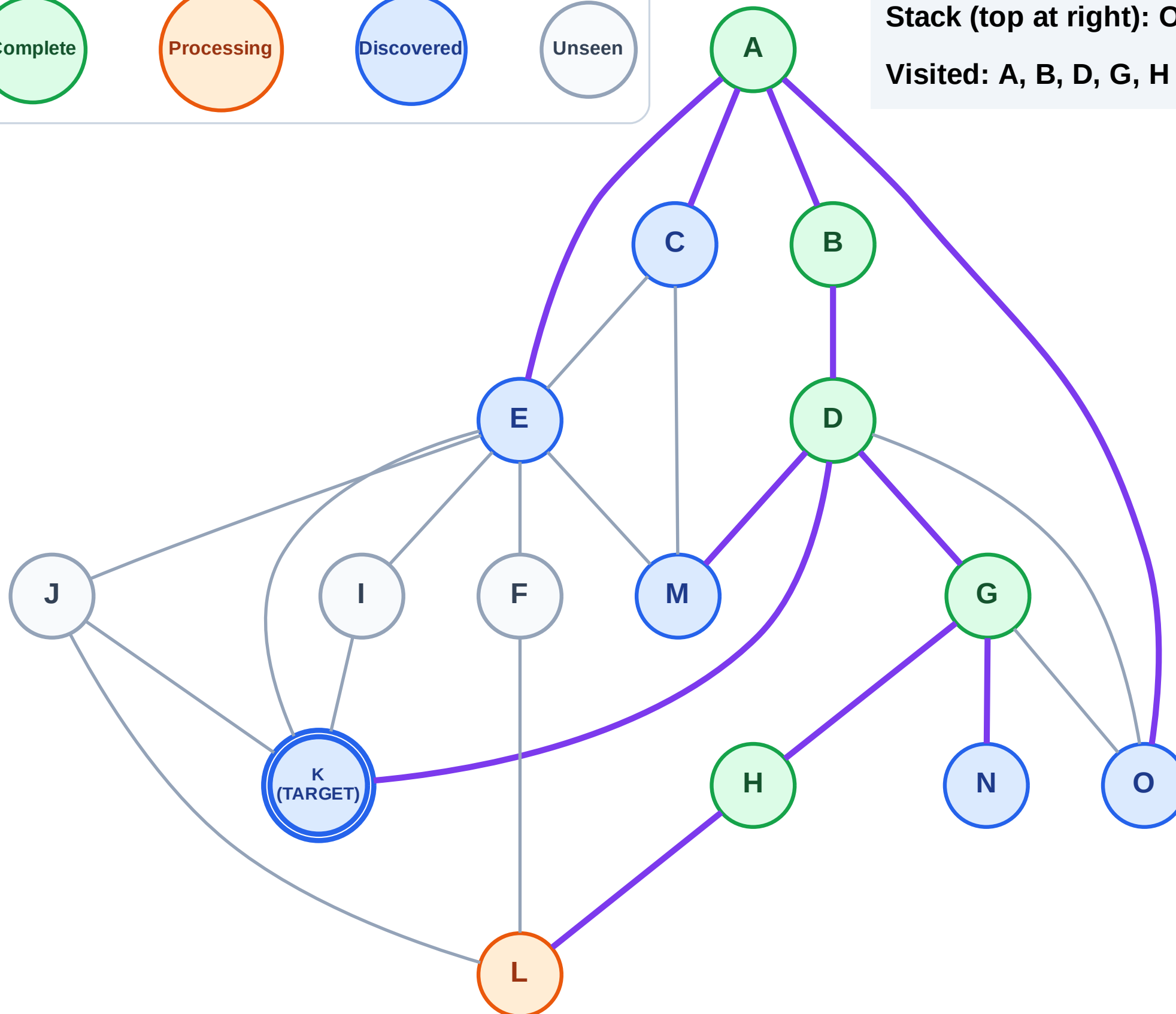
Step 12: Process node L

Legend



Stack (top at right): O | E | C | M | K | N

Visited: A, B, D, G, H



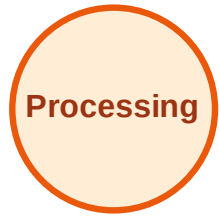
DFS Traversal

Step 13: Discover J, F from L

Legend



Complete



Processing



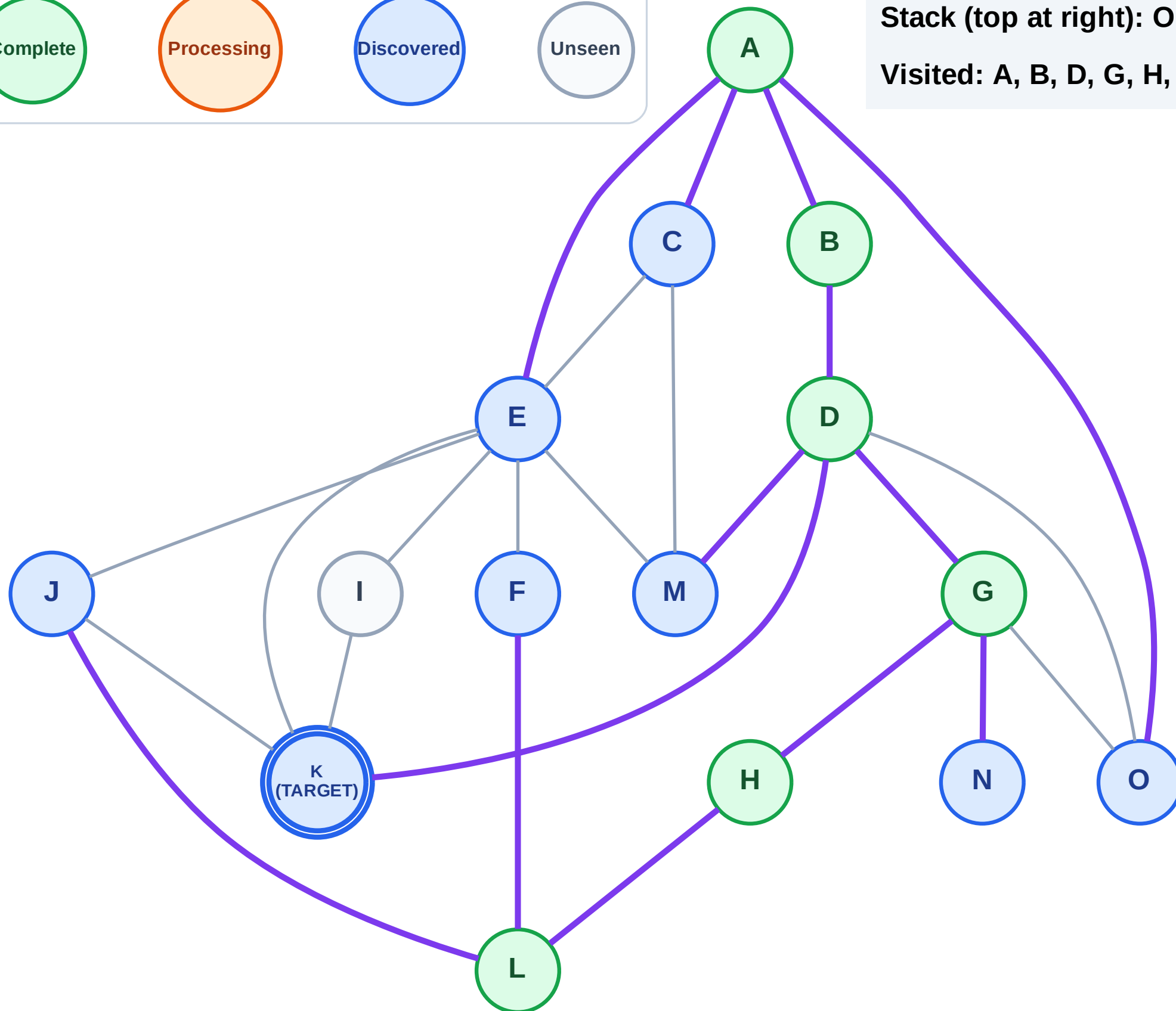
Discovered



Unseen

Stack (top at right): O | E | C | M | K | N | J | F

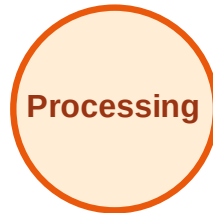
Visited: A, B, D, G, H, L



DFS Traversal

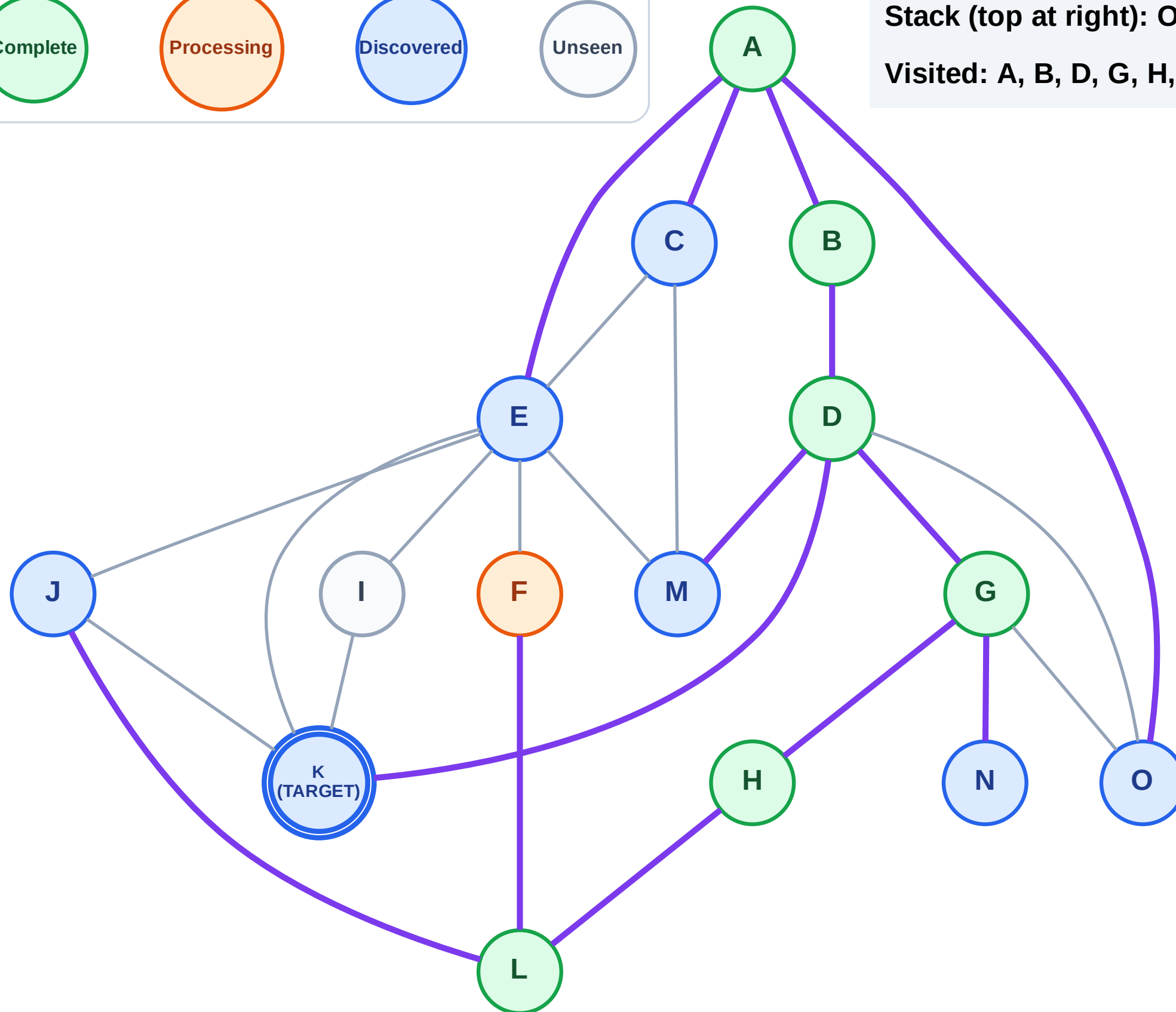
Step 14: Process node F

Legend



Stack (top at right): O | E | C | M | K | N | J

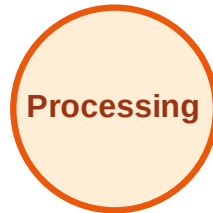
Visited: A, B, D, G, H, L



DFS Traversal

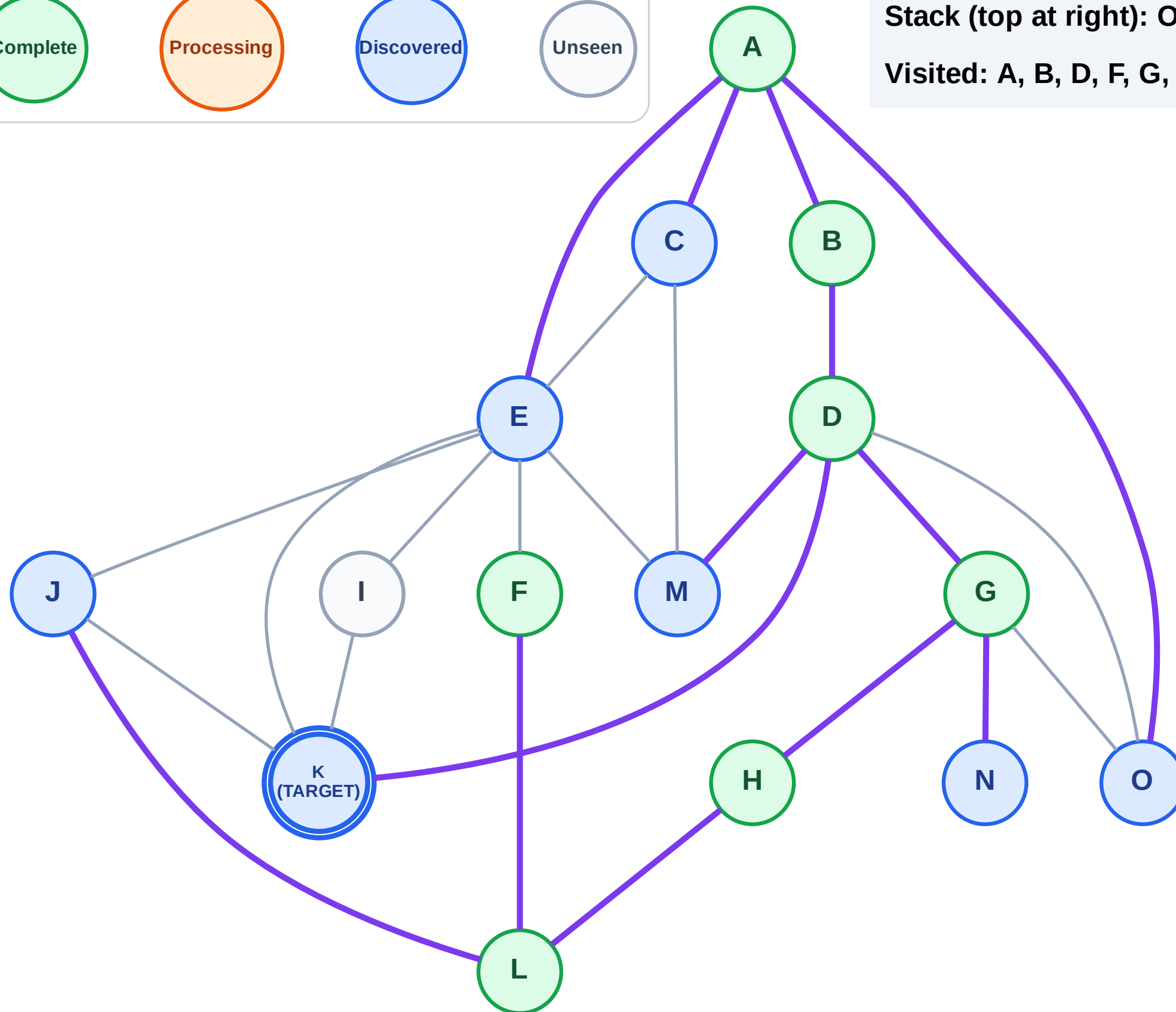
Step 15: No new neighbors from F

Legend



Stack (top at right): O | E | C | M | K | N | J

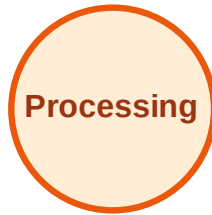
Visited: A, B, D, F, G, H, L



DFS Traversal

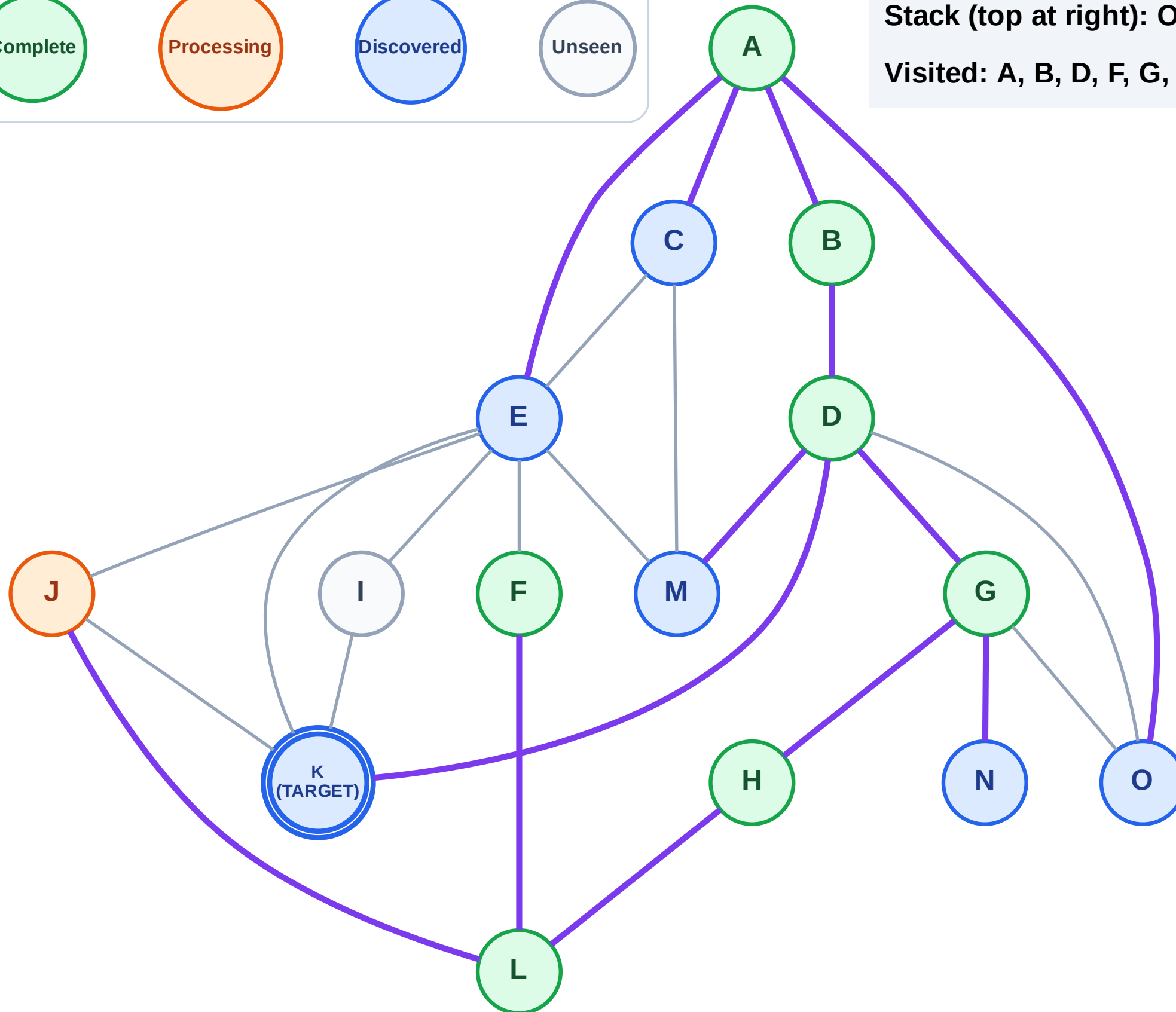
Step 16: Process node J

Legend



Stack (top at right): O | E | C | M | K | N

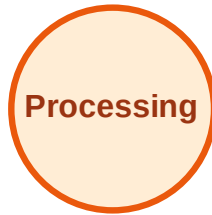
Visited: A, B, D, F, G, H, L



DFS Traversal

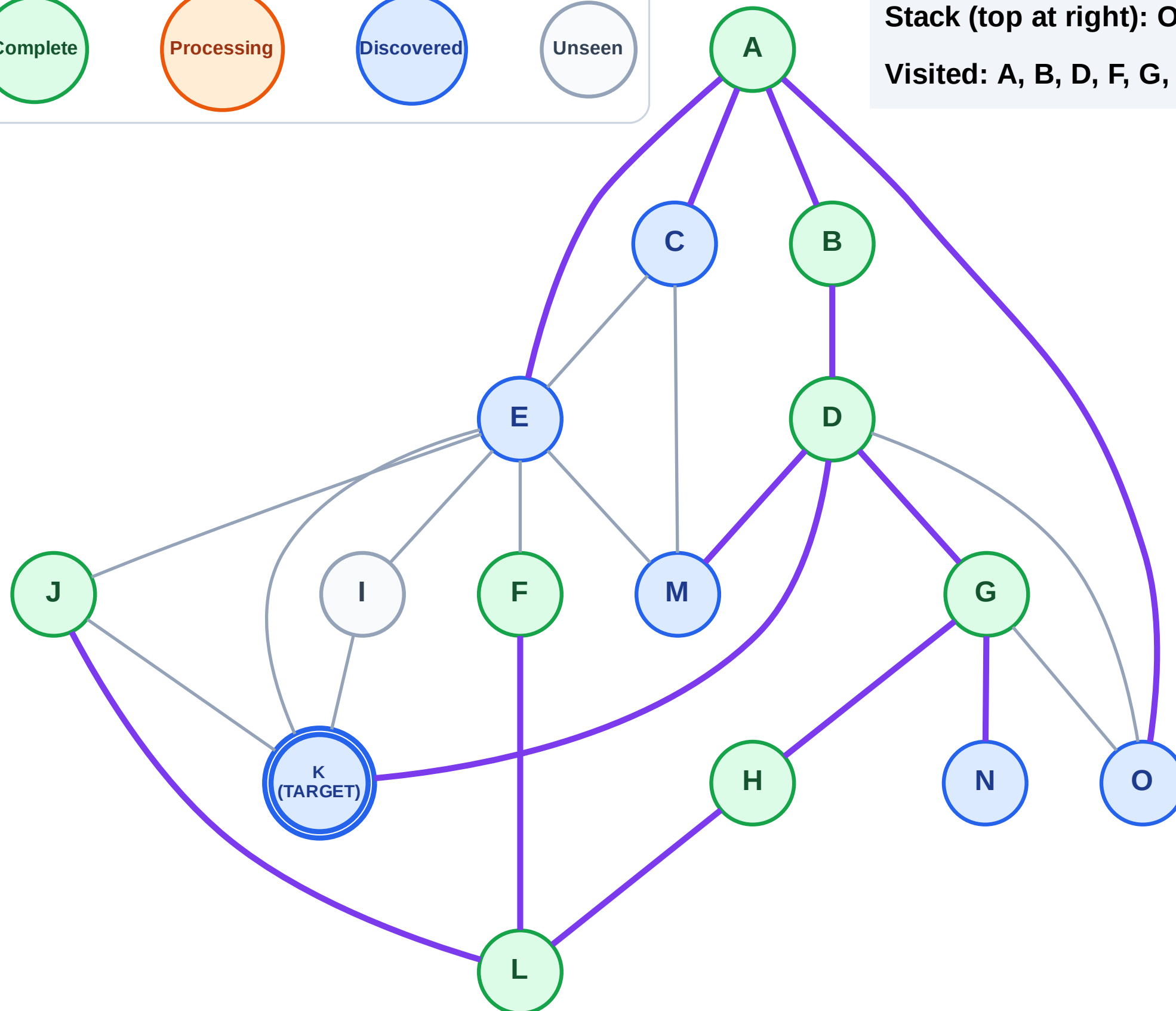
Step 17: No new neighbors from J

Legend



Stack (top at right): O | E | C | M | K | N

Visited: A, B, D, F, G, H, J, L



DFS Traversal

Step 18: Process node N

Legend

Complete

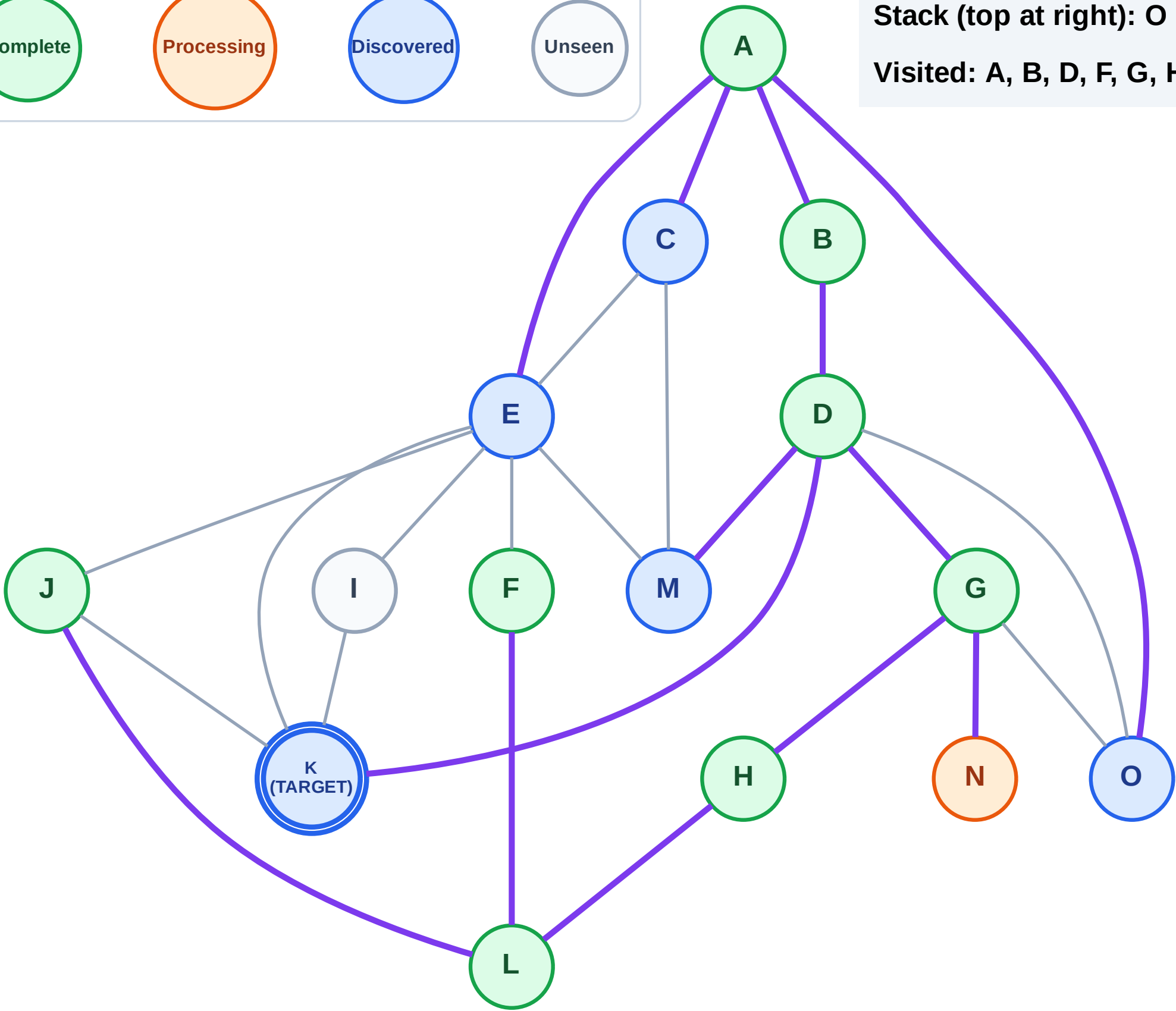
Processing

Discovered

Unseen

Stack (top at right): O | E | C | M | K

Visited: A, B, D, F, G, H, J, L



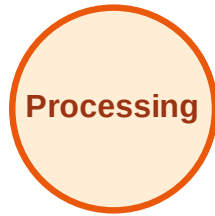
DFS Traversal

Step 19: No new neighbors from N

Legend



Complete



Processing



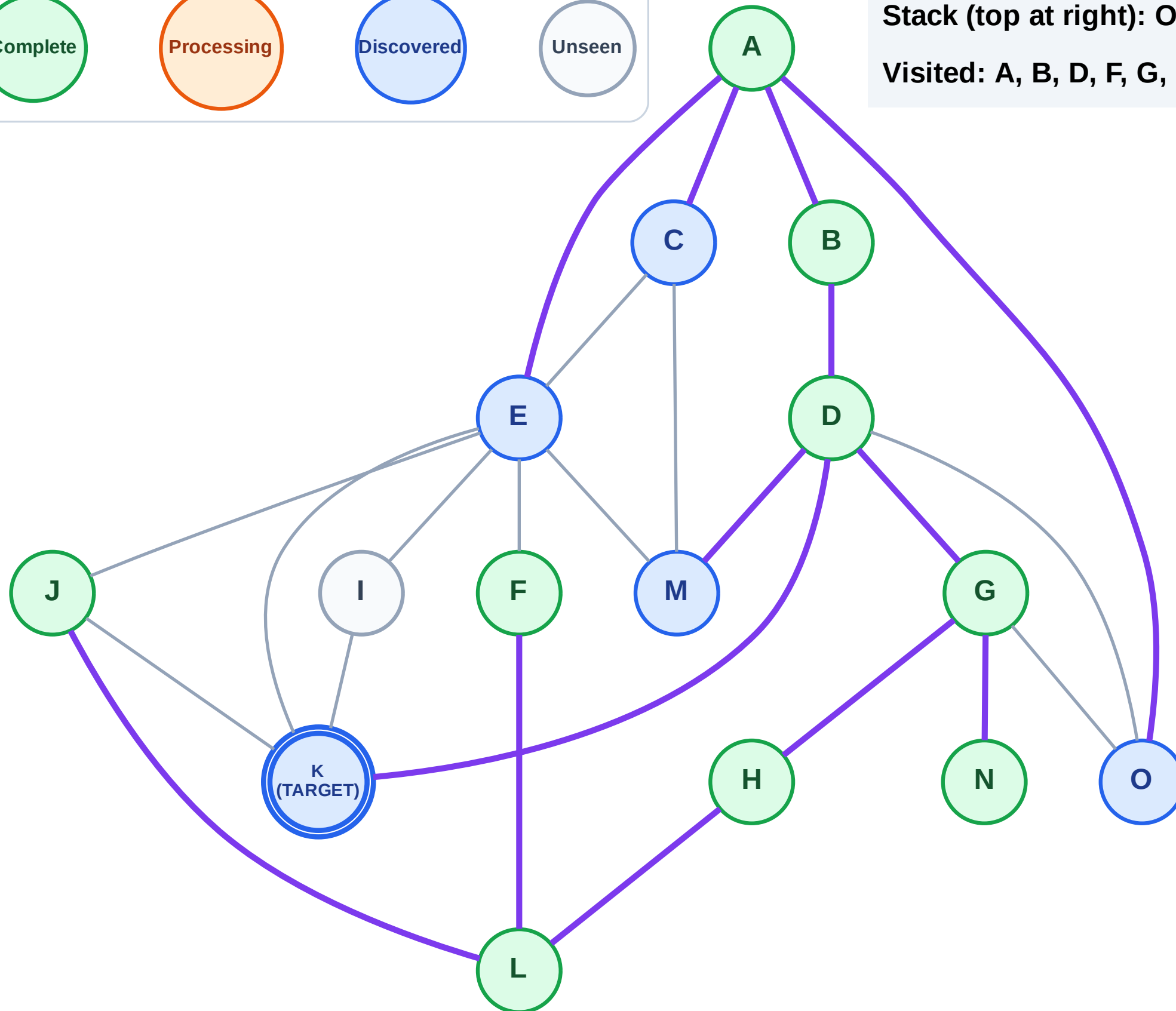
Discovered



Unseen

Stack (top at right): O | E | C | M | K

Visited: A, B, D, F, G, H, J, L, N



DFS Traversal

Step 20: Process node K

Legend

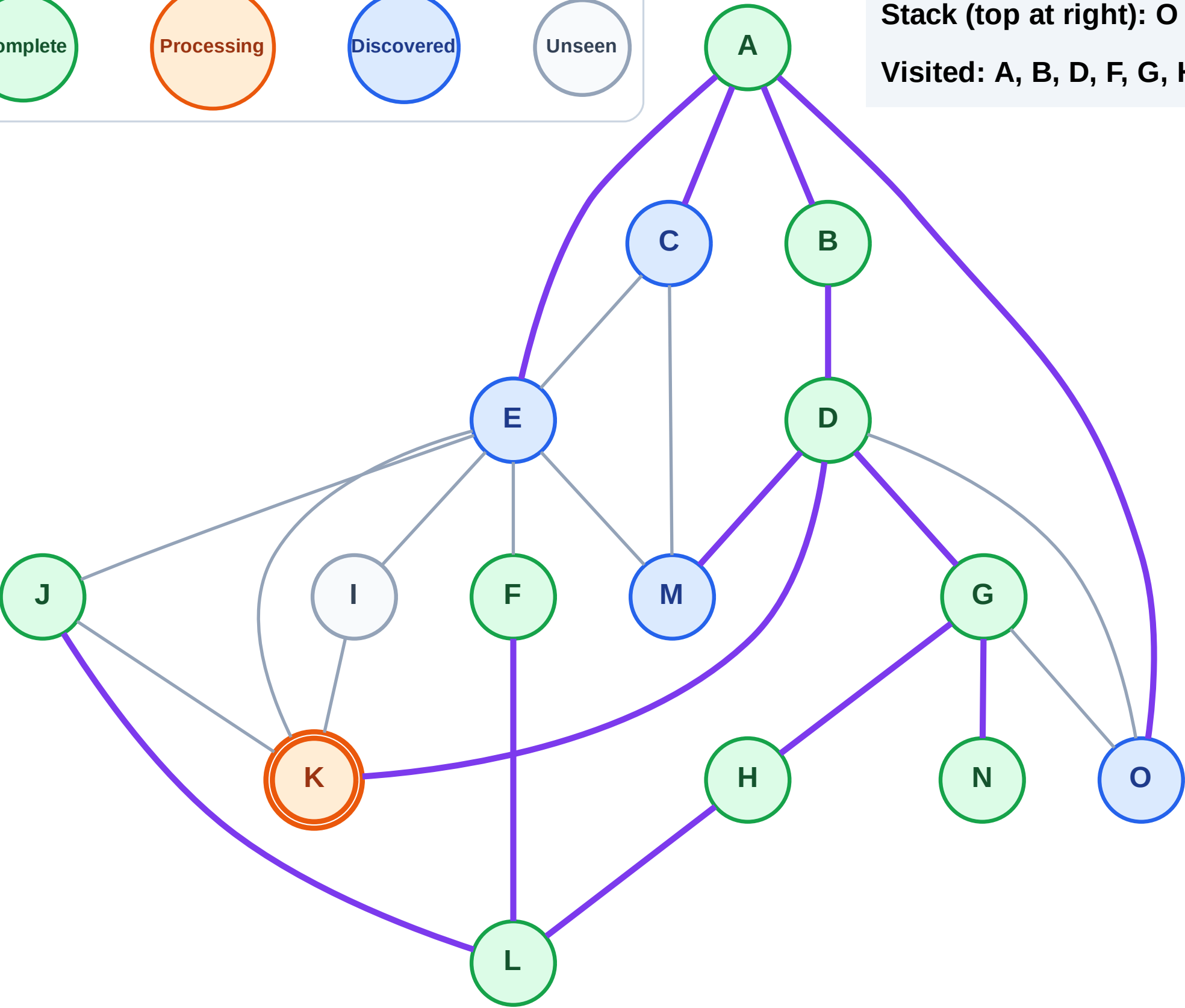
Complete

Processing

Discovered

Unseen

Stack (top at right): O | E | C | M
Visited: A, B, D, F, G, H, J, L, N



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (3 hops): A → B → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | E | C | M
Visited: A, B, D, F, G, H, J, K, L, N

